# HWs

Monday, 16 January 2023 4:16 PM

HW1

Saturday, 14 January 2023 9:24 PM

## Problem 2.1

Linearly independent reactions → No one stoichiometric equation can be derived from the others by a linear combination

(c) $A_i \rightarrow A_j$ ; $i, j = 1, 2, \ldots, n$, $i \neq j$

Linearly Ind. set:

$A_1 \rightarrow A_2$

$A_2 \rightarrow A_3$

$A_3 \rightarrow A_4$

$A_4 \rightarrow A_5$

⋮

$(n-1)$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & -1 & 1 & \cdots & 0 \\ 0 & 0 & 0 & -1 & 1 & \cdots \\ \vdots & & & & \ddots & \\ & & & & -1 & 1 \end{bmatrix}$$ (n columns, n−1 rows)

↳ $A_1 \rightarrow A_1$ is not possible (since $i \neq j$)

↳ $A_1 \rightarrow A_2$, $A_2 \rightarrow A_1$ } These form linearly dependent set since differ by a factor of −1.

∴ (n−1) Linearly independent reactions as shown in the set.

Uniqueness: The given set is not unique since other linearly independent sets can also be constructed

1) $A_1 \rightarrow A_2$, $A_2 \rightarrow A_3$, ... $A_{n-1} \rightarrow A_n$

2) $A_n \rightarrow A_1$, $A_n \rightarrow A_2$, ... $A_n \rightarrow A_{n-1}$

(d) $A_1 = NaCl$

$A_2 = H_2SO_4$

$A_3 = Na_2SO_4$

$A_4 = HCl$

$A_5 = C$

$A_6 = Na_2S$

$A_7 = CO$

$A_8 = CaCO_3$

$A_9 = Na_2CO_3$

$A_{10} = CaS$

$$\begin{cases} 2A_1 + A_2 \rightarrow A_3 + 2A_4 \\ A_3 + 4A_5 \rightarrow A_6 + 4A_7 \\ A_6 + A_8 \rightarrow A_9 + A_{10} \end{cases}$$

$$\begin{bmatrix} -2 & 1 & 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -4 & 1 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 1 & 1 \end{bmatrix}$$

rank = 3

10 reactants & 3 reactions

↳ Full rank matrix since no two equations can form the other

↳ 3 independent reactions and all three form a unique set

m × n

8 × 9

(e) $A_1 = C_3H_8$

$A_2 = C_2H_4$

$A_3 = CH_4$

$A_4 = C_3H_6$

$A_5 = H_2$

$A_6 = C_2H_6$

$A_7 = C_2H_2$

$A_8 = C_4H_8$

$A_9 = C_4H_6$

$A_1 \rightarrow A_2 + A_3$

$A_1 \rightarrow A_4 + A_5$

$A_1 + A_2 \rightarrow A_6 + A_4$

$2A_4 \rightarrow 3A_2$

$A_4 \rightarrow A_7 + A_3$

$A_4 + A_6 \rightarrow A_8 + A_3$

$A_6 \rightarrow A_2 + A_5$

$A_2 + A_7 \rightarrow A_9$

9 reactants, 8 reactions

$$\begin{bmatrix} -1 & +1 & +1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & +1 & +1 & 0 & 0 & 0 \\ -1 & -1 & 0 & +1 & 0 & +1 & 0 & 0 \\ 0 & +3 & 0 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & +1 & +1 & 0 & 0 & +1 & 0 \\ 0 & 0 & +1 & -1 & 0 & -1 & 0 & +1 \\ 0 & +1 & 0 & 0 & +1 & -1 & 0 & 0 \\ 0 & +1 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \\ A_5 \\ A_6 \\ A_7 \\ A_8 \\ A_9 \end{bmatrix}$$

rank = 7

(i.e. 7 pivots)

∴ 7 linearly independent reactions

Set:

All reactions excluding reaction 7

∴ $A_1 \rightarrow A_2 + A_3$

$A_1 \rightarrow A_4 + A_5$

$A_1 + A_2 \rightarrow A_6 + A_4$

$2A_4 \rightarrow 3A_2$

$A_4 \rightarrow A_7 + A_3$

$A_4 + A_6 \rightarrow A_8 + A_3$

$A_2 + A_7 \rightarrow A_9$

Uniqueness: The given set is not unique since one of reaction is linearly dependent and therefore can be expressed in a different set.

## Problem 2.3

Matrix A → stoichiometric matrix

Differential equation of Multiple reactions in batch

$$\frac{d\xi_i}{dt} = r_i - \xi_i \frac{d\ln V}{dt} \quad \text{... Eq 1}$$

If $\text{rank}(A) = r < m$ (m: no. of reactions)

∴ There are r linearly independent reactions given by the above equation (Eq 1) where $i = 1, 2, 3 \ldots, r$

Additionally, since volume is changing there is additional volume equation given by:

$$\frac{d\ln V}{dt} = \sum_{i=1}^{m} r_i \sum_{j=1}^{s} \alpha_j \bar{V}_j$$

∴ Minimum no. of differential eq: $r+1$

HW2

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Aq phase: $Q^+X^- + A^- \underset{k_2}{\overset{k_1}{\rightleftharpoons}} Q^+A^- + X^-$

Organic phase: $Q^+X^- + AB \overset{k_3}{\longleftarrow} Q^+A^- + BX$

Catalyst: $QX$
Complex: $QA$
Products: $AB$, $X^-$

Overall: $A^- + BX \longrightarrow AB + X^-$

$\frac{k_2}{k_1} = K \equiv$ equilibrium constant $= \frac{[QA]_a[X^-]}{[QX]_a[A^-]}$ ; Well-stirred reactor: uniform conc. of all species can be assumed where interphase conc. is equal to the bulk conc.

$[QA]_a = [QA]_{a,e}$ ; $[QX]_a = [QX]_{a,e}$

$[QA]_a = [QA]_{o,e}$ ; $[QX]_o = [QX]_{o,e}$

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$x = (1-\varphi)\frac{[QA]_o}{Q_T}$

(i) [a] Show $\frac{dU}{d\tau} = -UV$

In organic phase (for constant volume): $\frac{dc_j}{dt} = \alpha_j r$ ; $QA + BX \xrightarrow{k_3} QX + AB$

Assuming elementary reaction: $r_o = -k_3[QA]_o[BX]$

$\therefore \frac{d[BX]}{dt} = -k_3[QA]_o[BX]$ ; $\alpha_j = 1$

$\div$ by $[BX]_i$

$\frac{1}{[BX]_i}\frac{d[BX]}{dt} = -k_3\frac{[BX]}{[BX]_i}[QA]_o$

$\times \frac{\varphi}{Q_T}$

$\frac{\varphi}{Q_T\cdot[BX]_i}\frac{d[BX]}{dt} = -k_3\frac{[BX]}{[BX]_i}[QA]_o\cdot\varphi$

$\frac{d\left(\frac{[BX]}{[BX]_i}\right)}{d\left(\frac{t\cdot k_3\cdot Q_T}{\varphi}\right)} = -\frac{[BX]}{[BX]_i}\frac{\varphi[QA]_o}{Q_T}$

$$\boxed{\frac{dU}{d\tau} = -U\cdot V}$$

[b] Show: $\frac{dV}{d\tau} = -P_1(V - P_3X) - P_2UV$ ; $P_1 \equiv \frac{k_L a}{k_3 Q_T}$ ; $P_2 = \varphi\frac{[BX]_i}{Q_T}$ ; $P_3 = \pi_{QA}\frac{\varphi}{1-\varphi} = \frac{[QA]_{o,e}}{[QA]_{a,e}}$

$$\frac{d[QA]_o}{dt} = -k_3[QA]_o[BX] + K_L A([QA]_{o,e} - [QA]_o) \quad ; \quad \varphi = \frac{V_o}{V_T} \text{ (dispersed phase fraction)} \quad ; \quad V_T = V_o + V_a$$

$\frac{d[QA]_o}{dt}$: change in conc. of $[QA]_o$ — $-k_3[QA]_o[BX]$: Reaction in organic layer — $K_L A([QA]_{o,e} - [QA]_o)$: transport of $[QA]_o$ from aq. layer

; $A$ = interfacial area (effective mass transfer area)

$\left(\times \frac{\varphi}{k_3 \cdot Q_T}\right)$

$$\frac{d}{dt}\left(\frac{\varphi}{Q_T}[QA]_o\right) = -k_3[QA]_o[BX]\frac{\varphi}{Q_T} + \frac{\varphi}{Q_T}\frac{K_L A}{k_3}([QA]_{o,e} - [QA]_o)$$

$$\Rightarrow \frac{1}{k_3}\frac{dV}{dt} = -[QA]_o\frac{[BX]}{[BX]_i}\cdot\left([BX]_i\frac{\varphi}{Q_T}\right) + \frac{K_L(\varphi A)}{k_3 Q_T}([QA]_{o,e} - [QA]_o) \quad ; \quad \varphi A = a$$

$\left(\times \frac{\varphi}{Q_T}\right)$

$$\Rightarrow \frac{1}{\frac{k_3 Q_T}{\varphi}}\frac{dV}{dt} = -U\cdot P_2\left([QA]_o\frac{\varphi}{Q_T}\right) + P_1\left(\frac{\varphi}{Q_T}[QA]_{o,e} - \frac{\varphi}{Q_T}[QA]_o\right)$$

$$\frac{d\bar{V}}{d\tau} = -P_2 U\bar{V} + P_1\left(\frac{\varphi}{Q_T}\frac{[QA]_{o,e}}{[QA]_{a,e}}\cdot[QA]_a - \bar{V}\right) \quad ; \quad [QA]_{a,e} = [QA]_a$$

$$\frac{d\bar{V}}{d\tau} = -P_2 U\bar{V} + P_1\left(\frac{\varphi}{(1-\varphi)}\pi_{QA}\cdot\frac{(1-\varphi)}{Q_T}[QA]_a - \bar{V}\right)$$

$$\frac{d\bar{V}}{d\tau} = -P_2 U\bar{V} + P_1\left(P_3\cdot X - \bar{V}\right)$$

$$\therefore \boxed{\frac{d\bar{V}}{d\tau} = -P_2 U\bar{V} - P_1(\bar{V} - P_3 X)}$$

[c] Show: $\frac{dW}{d\tau} = -P_1(W - P_4 Y) + P_2 UV$ ; $P_4 = \pi_{QX}\frac{\varphi}{1-\varphi}$

Similar to previous part: $\frac{d[QX]_o}{dt} = k_3[QA]_o[BX] - K_L A([QX]_o - [QX]_{o,e})$

$\frac{d[QX]_o}{dt}$: Rate of change of $[QX]$ — $k_3[QA]_o[BX]$: rate at which produced in organic phase — $K_L A([QX]_o - [QX]_{o,e})$: transport of $[QX]$ to aq. layer

$\left(\times \frac{\varphi^2}{k_3 Q_T^2}\right)$:

$$\frac{d\left(\frac{[QX]_o\varphi}{Q_T}\right)}{d\left(\frac{t\cdot k_3\cdot Q_T}{\varphi}\right)} = \varphi\frac{[BX]_i}{Q_T}\cdot\frac{[BX]}{[BX]_i}\cdot\frac{[QA]_o\varphi}{Q_T} - \frac{K_L(A\cdot\varphi)}{Q_T k_3}\left(\frac{[QX]_o\varphi}{Q_T} - \frac{[QX]_{o,e}}{[QX]_{a,e}}\cdot[QX]_a\cdot\frac{\varphi}{Q_T}\cdot\frac{\varphi}{(1-\varphi)}(1-\varphi)\right)$$

$$\Rightarrow \boxed{\frac{dW}{d\tau} = +P_2 UV - \frac{k_L a}{Q_T k_3}\left(W - P_4 Y\right)}$$

[d] Show: $1 = V + W + X + Y$

Mole balance on catalyst ($Q^+$) ; $C_j = \frac{N_j}{V}$ ; $V_T = V_o + V_a$

$$V_a \cdot [QX]_a + [QA]_a V_a + [QA]_o V_o + [QX]_o \cdot V_o = Q_T \cdot V_T$$

$$\Rightarrow \frac{V_a}{V_T}\frac{[QX]_a}{Q_T} + \frac{V_a}{V_T}\frac{[QA]_a}{Q_T} + \frac{V_o}{V_T}\frac{[QA]_o}{Q_T} + \frac{V_o}{V_T}\frac{[QX]_o}{Q_T} = 1$$

$$\Rightarrow \frac{(V_T - V_o)}{V_T}\frac{[QX]_a}{Q_T} + \frac{(V_T - V_o)}{V_T}\frac{[QA]_a}{Q_T} + \Phi\frac{[QA]_o}{Q_T} + \Phi\frac{[QX]_o}{Q_T} = 1$$

$$\Rightarrow (1-\Phi)\frac{[QX]_a}{Q_T} + (1-\Phi)\frac{[QA]_a}{Q_T} + V + W = 1$$

$$\therefore \boxed{Y + X + V + W = 1}$$

(e) Show $P_5(1-Z) = P_2(1-U) + V + X$ ; $Z = \frac{[A^-]}{[A^-]_i}$ ; $P_5 = (1-\Phi)\frac{[A^-]_i}{Q_T}$

Overall reaction: $A^- + BX \rightarrow AB + X^-$

Mole balance on $A^-$ species:

$$\Delta N_{A^-} = N_{QA,a} + N_{QA,o} + N_{AB}$$

Converting to concentrations,

$$V_a([A]_i - [A]) = V_a[QA]_a + V_o[QA]_o + [AB]\cdot V_o$$

$$V_a([A]_i - [A]) = V_a[QA]_a + V_o[QA]_o + ([BX]_i - [BX])V_o$$

(Starting w/ no initial AB)

$$\xi = \frac{\cancel{[AB]_i}^{\,0} - [AB]}{1} = \frac{[BX]_i - [BX]}{-1}$$

$$\xi = [AB] = [BX]_i - [BX]$$

Dividing by total volume and $Q_T$:

$$\Rightarrow \frac{V_a}{V_T \cdot Q_T}([A]_i - [A]) = \frac{V_a}{V_T Q_T}[QA]_a + \frac{V_o}{V_T Q_T}[QA]_o + \frac{V_o}{V_T Q_T}([BX]_i - [BX])$$

$$\Rightarrow (1-\Phi)\frac{[A]_i}{Q_T}\left(1 - \frac{[A]}{[A]_i}\right) = \frac{[BX]_i}{Q_T}\cdot\Phi\cdot\left(1 - \frac{[BX]}{[BX]_i}\right) + \frac{(1-\Phi)}{Q_T}[QA]_a + \frac{\Phi}{Q_T}[QA]_o$$

$$\Rightarrow \boxed{P_5(1-Z) = P_2(1-U) + X + V}$$

(f) Show: $KP_5YZ = X[P_2(1-V) + 1 - W - Y]$

Mole balance on $[X^-]$: ; $K = \frac{[QA]_a[X^-]}{[A^-][QX]_a} = \frac{k_1}{k_2}$ ; $\xi = \frac{[X^-]_i - [X^-]}{1} = \frac{[A^-]_i - [A^-]}{-1}$

$$N_{X^-,i} = N_{X^-,f}$$

$$[BX]_i \cdot V_0 + [X^-]_i V_a + [QX]_i \cdot V_a = V_0 \cdot [BX] + V_0 [QX]_0 + V_a \cdot [X^-] + V_a \cdot [QX]_a \quad ; \text{ Assuming all initial } [QX]_i = [QX]_{a,i} = Q_T$$

Dividing by $V_T$:

$$\Rightarrow (1-\Phi)(\cancel{[X]_i^0} - [X^-]) + \Phi([BX]_i - [BX]) = \Phi[QX]_0 + (1-\Phi)([QX]_a - Q_T)$$

$$\Rightarrow -(1-\Phi)[X^-] + \Phi[BX]_i\left(1 - \frac{[BX]}{[BX]_i}\right) = \Phi[QX]_0 - (1-\Phi)Q_T + (1-\Phi)[QX]_a$$

$$\times \frac{(1-\Phi)[QA]_a}{Q_T^2}$$

$$\Rightarrow \frac{(1-\Phi)^2[QA]_a[X^-]}{Q_T^2} = -\frac{\Phi(1-\Phi)[QX]_0[QA]_a}{Q_T^2} - \frac{(1-\Phi)^2[QA]_a[QX]_a}{Q_T^2} + \frac{(1-\Phi)[QA]_a}{Q_T} + \frac{(1-\Phi)\Phi[QA]_a[BX]_i}{Q_T^2}\left(1 - \frac{[BX]}{[BX]_i}\right)$$

$$= \frac{(1-\Phi)[QA]_a}{Q_T}\left[1 + \frac{\Phi}{Q_T}[BX^-](1-U) - \frac{\Phi[QX]_0}{Q_T} - \frac{(1-\Phi)[QX]_a}{Q_T}\right]$$

$$\Rightarrow K\frac{(1-\Phi)[A^-]_i}{Q_T}\frac{(1-\Phi)[QX]_a}{Q_T}\frac{[A^-]}{[A^-]_i} = \frac{(1-\Phi)[QA]_a}{Q_T}\left[\frac{\Phi[BX]_i}{Q_T}\left(1 - \frac{[BX]}{[BX]_i}\right) + 1 - \frac{\Phi[QX]_0}{Q_T} - \frac{(1-\Phi)[QX]_a}{Q_T}\right)$$

$$\therefore \boxed{KP_5YZ = X[P_2(1-U) + 1 - W - Y]}$$

Rest of HW completed on Python (see attached)

HW3

Tuesday, 31 January 2023 12:01 PM

## Problem 6.1

Kinetic expression : + Single substrate (S)
+ Product (P)
+ Inhibitory effect ( E-S complex can add on P for an E-S-P complex in a reversible reaction)
+ sketch (rate of product as function of substrate conc.)

Overall: $E + S \rightarrow E + P$

Michaelis-Menton: $E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES$ $\qquad ES \xrightarrow{k_2} E + P$

Product Inhibition: $ES + P \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} ESP$

@ equilibrium: $[ESP] = K_3[ES][P]$ ; $K_3 = \frac{k_3}{k_{-3}}$

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] = k_1[E][S] - [ES](k_{-1} + k_2)$$

$$\frac{d[P]}{dt} = k_2[ES]$$

For equilibrium reaction for step 1: $K_1 = \frac{[ES]}{[E][S]}$

$$[ES] = K_1[E][S]$$

Total conc. of enzyme sites [Site balance]:
$$[E]_0 = [E] + [ES] + [ESP]$$
$$= [E] + [ES] + K_3[ES][P]$$
$$= [E] + [ES](1 + K_3[P])$$

Substituting values for [ES],

$$[E]_0 = [E] + K_1[E][S](1 + K_3[P])$$
$$[E]_0 = [E]\{1 + K_1[S] + K_1K_3[S][P]\} \Rightarrow [E] = \frac{[E]_0}{(1 + K_1[S] + K_1K_3[S][P])}$$

Rate of product formation: $V = \frac{d[P]}{dt} = k_2[ES] - k_3[P][ES] + k_{-3}[ESP] = k_2[ES] - k_3[P][ES] + k_{-3}K_3[ES][P] = k_2[ES] + [ES][P](-k_3 + k_{-3}K_3) = k_2[ES]$ (the $-k_3 + k_{-3}K_3$ term $\rightarrow 0$)

$$\frac{d[P]}{dt} = k_2K_1[E][S] = \boxed{\frac{k_2K_1[S][E]_0}{1 + K_1[S] + K_1K_3[S][P]} = V}$$

We have to plot V vs [S]

We can find [S] as a function of [P]:

Balance on [E] and [P]: (You need 2 because one moles of S & P gives 1 mole of ESP)

$$[S]_0 = [S] + [ES] + 2[ESP] + [P]$$
$$[S]_0 = [S] + [ES] + 2(K_3[ES][P]) + [P]$$
$$[S]_0 = [S] + (K_1[E][S])(1 + 2K_3[P]) + [P]$$

Solving for [P]:
$$\boxed{[P] = \frac{[S]_0 - [S] - K_1[E][S]}{(1 + 2K_1K_3[E][S])}}$$

This is also an implicit function

Therefore combining expressions of [E], [P] and iterating [S]

Now we can iterate values for [S] and knowing constants $K_1, k_2, K_3$ we can plot V $[S]_0$. attached

## Question 6.2

$P \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 1P$

Question 6.2

Hill expression ; L = ligand ; Protein P ; $L + P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} LP$

$$\frac{d[LP]}{dt} = k_1[L][P] - 1[LP]$$

(i) Fraction of bound protein to total protein by $\theta$:

@ Equilibrium: $\frac{1}{K} = \frac{[LP]}{[L][P]}$ ; $[LP] = \frac{1}{K}[L][P]$ ; $K = \frac{k_2}{k_1}$

(or $\frac{d[LP]}{dt} = 0$ @ss)

Total protein: $[P]_0 = [P] + [LP] = [P] + \frac{1}{K}[L][P]$

$$\theta = \frac{[LP]}{[P]_0} = \frac{\frac{1}{K}[L][P]}{[P] + \frac{1}{K}[L][P]} = \frac{\frac{1}{K}[L]}{\left(1 + \frac{1}{K}[L]\right)} = \boxed{\frac{[L]}{K + [L]} = \theta}$$

(ii) $P + 2L \xrightarrow{k_1} LP + L$ ; $LP \xrightarrow{k_2} L + P$

$$\frac{d[LP]}{dt} = k_1[P][L]^2 - k_2[LP]$$ ; Total protein: $[P]_0 = [P] + [LP]$

$$= [P] + \frac{k_1}{k_2}[P][L]^2 = [P] + \frac{1}{K}[P][L]^2$$

$\frac{d[LP]}{dt} = 0$ → $[LP] = \frac{k_1}{k_2}[P][L]^2 = \frac{1}{K}[P][L]^2$

@ss

$$\theta = \frac{[LP]}{[P]_0} = \frac{\frac{1}{K}[P][L]^2}{[P] + \frac{1}{K}[P][L]^2} = \boxed{\frac{[L]^2}{K + [L]^2} = \theta}$$

(iii) Based on part (ii), I would assume $P + nL \rightarrow LP + (n-1)L$ ; $LP \rightarrow L + P$

(iv) Scale invariance: $aA + bB \rightarrow C$ ; $\frac{r(\alpha[A], \beta[B])}{r([A], [B])} = \phi(\alpha, \beta)$

↓ Do not change if other variables

↑ This ratio of rates of arbitrary exponent values over the stoichiometric rate expressions

Properties of $\phi(\alpha, \beta)$: a) $\phi = 1$, when the rate expressions for stoichiometric elementary mechanisms match that of $\alpha$, $\beta$ expression

$r([A], [B]) = [A]^a[B]^b$

$r(\alpha[A], \beta[B]) = (\alpha[A])^a(\beta[B])^b = \alpha^a[A]^a\beta^b[B]^b = \alpha^a\beta^b[A]^a[B]^b = \alpha^a\beta^b r([A], [B])$

$$\therefore \boxed{\frac{r(\alpha[A], \beta[B])}{r([A], [B])} = \alpha^a\beta^b = f(\alpha, \beta) = \phi(\alpha, \beta)}$$

(v) $\frac{\partial r(\alpha[A], \beta[B])}{\partial \alpha} = \frac{\partial}{\partial \alpha}\left[\alpha^a\beta^b \cdot r([A], [B])\right]$ ; from part (iv)

$= a\alpha^{a-1}\beta^b r([A], [B])$ ; holding $[A]$, $[B]$, $\beta$ constant

$= a\frac{\alpha^a}{\alpha}\beta^b r([A], [B])$

We also know $\phi = \alpha^a\beta^b$ ; $\frac{\partial \phi}{\partial \alpha} = a\alpha^{a-1}\beta^b = a\frac{\alpha^a}{\alpha}\beta^b r([A], [B])$

$$\therefore \boxed{\frac{[A]}{[A]}\frac{\partial r(\alpha[A],\beta[B])}{\partial \alpha} = r([A],[B])\cdot\frac{\partial \phi(\alpha,\beta)}{\partial \alpha}}$$

$\rightarrow$ $x=\alpha[A]$ ; $y=\beta[B]$ :

$$[A]\frac{\partial r(x,y)}{\partial([A]\cdot\alpha)} = r([A],[B]\cdot\beta)\frac{\partial\phi(\alpha,\beta)}{\partial\alpha} \quad ; \quad \beta=1$$

$$[A]\frac{\partial r(x,y)}{\partial x} = r([A],y)\frac{\partial\phi(\alpha,1)}{\partial\alpha}$$

$\rightarrow$ $\alpha=1$ :

$x=\alpha[A]=[A]$ :

$$x\cdot\frac{\partial r(x,y)}{\partial x} = r(x,y)\frac{\partial\phi(1,1)}{\partial\alpha}$$

If we let $\mu \equiv \frac{\partial\phi(1,1)}{\partial\alpha}$

$$\Rightarrow x\frac{\partial r}{\partial x} = r\cdot\mu \rightarrow \text{This can be solved as an ODE of order 1}$$

$$\int\frac{dr}{r} = \int\frac{\mu}{x}\cdot dx$$

$$\ln r = \mu\ln x + c$$
$$\ln r = \ln x^{\mu} + c \quad ; \; c \text{ has to be a function } y \text{ since we integrated w.r.t to } x$$
$$\ln r = \ln x^{\mu} + f(y)$$

$f(y)$ can be expressed as $\ln[f(y)] = \ln[\psi(y)]$

$$\Rightarrow \ln r = \ln[\psi(y)\cdot x^{\mu}] \quad \text{Taking exponentials of both sides.}$$

$$\therefore \boxed{r(x,y) = \psi(y)x^{\mu}}$$

(vi) Following the above steps but w/ respect to $\beta$:

$$\Rightarrow \quad r(\alpha[A],\beta[B]) = r([A],[B])\cdot\alpha^a\beta^b$$

$$\Rightarrow \quad \frac{\partial r(\alpha[A],\beta[B])}{\partial\beta} = r([A],[B])\frac{\partial}{\partial\beta}(\alpha^a\beta^b)$$

$\times\frac{[B]}{[B]}$

$$[B]\frac{\partial r(\alpha[A],\beta[B])}{\partial\beta[B]} = r([A],[B])\alpha^a\frac{b\beta^b}{\beta} \quad ; \quad \frac{\partial\phi}{\partial\beta} = \alpha^a\frac{b\beta^b}{\beta}$$

$$= r([A],[B])\frac{\partial\phi(\alpha,\beta)}{\partial\beta}$$

Setting $x=\alpha[A]$, $y=\beta[B]$

$$[B]\frac{\partial r(x,y)}{\partial y} = r([A],[B])\frac{\partial\phi(\alpha,\beta)}{\partial\beta}$$

Setting $\alpha=1$, $\beta=1$ :

$$y\frac{\partial r}{\partial y} = r(x,y)\frac{\partial\phi(1,1)}{\partial\beta}$$

Solving ODE:

$$y\frac{\partial r}{\partial y} = r\,\nu$$

$$\int \frac{\partial r}{r} = \int \nu \frac{\partial y}{y}$$

→ since $x$ was held constant

$$\ln r = \nu \ln y + \ln(\chi(x))$$

$$\boxed{r(x,y) = \chi(x)\, y^{\nu}}$$

(vii) Hill expression: $L + P \rightarrow LP$
[A] [B] [C]

As seen from above expressions, values of a & b as stoichiometric coefficients are absorbed by $\phi$ term and the final rate expressions have similar expressions regardless of the value of a & b proposed. Hence those can be non-integer.

(i.e. $\alpha$ & $\beta$ can scale accordingly to obtain $r(x,y) = \lambda x^{\mu} y^{\nu}$ where $\mu = \frac{\alpha^a \beta^b}{\alpha}$ ; $\nu = \frac{\alpha^a \beta^b}{b}$ can be any non-integer value)

HW4

Tuesday, 7 February 2023 9:19 AM

## Question 10.2

$$\Phi^2 = \frac{-\phi^2 D^{-1} K}{A} = -\phi^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1000 \end{bmatrix}^{-1} \begin{bmatrix} -2 & 1 & 0 \\ 2 & -2 & 2 \\ 0 & 1 & -2 \end{bmatrix}$$

$N' = \Phi^{-1} \tanh \Phi$ → show in terms of $\phi$

Evaluating $\Phi^2$ expression: (using wolfram Alpha)

$$\Phi^2 = -\phi^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0.001 \end{bmatrix} \begin{bmatrix} -2 & 1 & 0 \\ 2 & -2 & 2 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\Phi^2 = -\phi^2 \begin{bmatrix} -2 & 1 & 0 \\ +2 & -2 & +2 \\ 0 & 0.001 & -0.002 \end{bmatrix}$$

From notes:

$$\Phi^{-1} \tanh \Phi = P_0 + \sum_{j=1}^{n-1} \lambda_j^{-1} \tanh(\lambda_j P_j) \quad ; \quad P_j \rightarrow \text{projection matrix}$$

$$P_j = \frac{u_j v_j^T}{v_j^T u_j} \qquad u \rightarrow \text{right eigenvector}, \quad v \rightarrow \text{left eigenvector}$$

$P_0 \rightarrow$ projection matrix corresponding to $\lambda = 0$

Find $\Phi = \sqrt{\Phi^2}$ from python:

$$\Phi = \begin{bmatrix} 1.31 & -0.383 & -0.54 \\ -0.766 & 1.306 & -1.845 \\ -5.4\times10^{-4} & -9.23\times10^{-4} & 2.386\times10^{-3} \end{bmatrix} \rightarrow \text{eigenvalues} \quad \begin{matrix} \lambda_1 = 1.8478 \\ \lambda_2 = 0.7665 \\ \lambda_3 = 0 \end{matrix}$$

Finding left and rite eigenvalues to construct the projection matrix:

$$N' = \Phi^{-1} \tanh \Phi = \begin{bmatrix} 0 & 0 & 0.997 \\ 0 & 0 & 1.994 \\ 0 & 0 & 0.997 \end{bmatrix} + \frac{\tanh(\lambda_1 \phi)}{\lambda_1} \begin{bmatrix} 0.499 & -0.35 & 0.207 \\ -0.707 & 0.5 & -0.293 \\ 0 & 0 & 0 \end{bmatrix} + \frac{\tan(\lambda_2 \phi)}{\lambda_2} \begin{bmatrix} 0.499 & 0.352 & -1.204 \\ 0.705 & 0.497 & -1.701 \\ 0 & 0 & 0 \end{bmatrix}$$

$\lambda_1 = 1.85$ $\qquad$ $\lambda_2 = 0.767$

(b) Selectivity to paraxylene; $\text{selectivity} = \frac{\text{desired product}}{\text{undesired product}}$

From class,

$$\text{reaction rate} = \hat{K} N' C_0 \quad ; \quad C_0 = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix} \quad ; \quad \hat{K} \rightarrow \text{rate constant matrix per unit area}$$

(for every $C_0$) ($C_0$: conc. vector at pore mouth)

$r_3$ = reaction rate of paraxylene described by [Third row of $kN'c_0$ matrix]

$r_2$ = reaction rate of Metaxylene [Second row of $kN'c_0$ matrix]

$r_1$ = reaction rate of Ortho [First row of $kN'c_0$ matrix]

$$\text{Selectivity} = \frac{r_3}{r_2 + r_1} = \frac{e_3^T \hat{K} N' c_0}{e_1^T \hat{K} N' c_0 + e_2^T \hat{K} N' c_0}$$

e vectors pick out the specific rows defining the rates

(C) Code solves for the various conditions → Attached

Analysis:

It seems the diffusion coefficient for para ($A_3$) is order of magnitude greater than those of ortho and meta. N' has inverse diffusion dependence but it is nested in the inverse and square roots

→ Selectivities of -1 means that any para formed is being consumed again to form different isomers.

Question 2 : GA

HW 5

Monday, 13 February 2023 4:58 PM

Problem 12.1

$$\hat{r}(c) = \frac{k \cdot c}{K' + Kc + c^2}$$

(i) Max reaction rate:

$$\frac{d\hat{r}}{dc} = \frac{(K' + Kc + c^2)\cdot k - k\cdot c\cdot(K + 2c)}{(K' + Kc + c^2)^2} = 0$$

$$\Rightarrow \frac{kK' + kKc + kc^2 - kKc - 2kc^2}{(K' + Kc + c^2)^2} = 0$$

$$\Rightarrow \frac{kK' - kc^2}{(K' + Kc + c^2)^2} = 0$$

$$kK' = kc^2$$

$$c = \sqrt{K'} \longrightarrow \text{Point of Local extrema [Taking positive value)}$$

Max? $\frac{d^2\hat{r}}{dc^2} = \frac{(K' + Kc + c^2)^{\cancel{2}}\cdot(-2kc) - 2(K' + Kc + c^2)\cdot(K + 2c)\cdot(kK' - kc^2)}{(K' + Kc + c^2)^{\cancel{4}3}}$ ; @ maximum critical point: $\frac{d^2\hat{r}}{dc^2} < 0$

$$= \frac{(K' + Kc + c^2)(-2kc) - 2(K + 2c)(kK' - kc^2)}{(K' + Kc + c^2)^3}$$

$$= \frac{-2kK'c - 2kKc^2 - 2kc^3 - 2kKK' + 2kKc^2 - 4kK'c + 4kc^3}{(K' + Kc + c^2)^3}$$

$$\frac{d^2\hat{r}}{dc^2} = \frac{2kc^2 - 6kK'c - 2kKK'}{(K' + Kc + c^2)^3}$$

@ $c = \sqrt{K'} \Rightarrow \frac{2kK' - 6kK'\sqrt{K'} - 2kKK'}{(K' + \sqrt{K}K + K')^3}$

K', k, K are defined as positive constants :→ $(K' + K\sqrt{K'} + K)^3 > 0$ [Denominator is always positive]

Numerator: $kK'\underbrace{(2 - 6\sqrt{K'} - 2K)}$

This has to be a negative value since K and K' cannot be less than zero

(ii) Plate type catalyst pellet (half thickness a): $u = \frac{c}{c_s}$ ; $c = u\cdot c_s$ ; $\hat{r}(c_s) = \frac{kc_s}{K' + Kc_s + c_s^2}$

$$R(u) = \frac{\hat{r}(c)}{\hat{r}(c_s)} = \frac{\frac{\cancel{k}c}{K' + Kc + c^2}}{\frac{\cancel{k}c_s}{K' + Kc_s + c_s^2}} = \frac{(c/c_s)}{(K' + Kc + c^2)}\cdot(K' + Kc_s + c_s^2)$$

$$= \frac{(c/c_s)\qquad (K' + Kc_s + c_s^2)}{\left[\frac{K'}{K' + Kc_s + c_s^2} + \frac{Kc}{K' + Kc_s + c_s^2} + \frac{c^2}{K' + Kc_s + c_s^2}\right](K' + Kc_s + c_s^2)}$$

↓ Taking $(K' + Kc_s + c_s^2)$ as common factor from denominator

↓ $u = \frac{c}{c_s}$

$$= \frac{u}{\left[\underbrace{\frac{K'}{K' + Kc_s + c_s^2}}_{\alpha} + \underbrace{\frac{Kc_s\,u}{K' + Kc_s + c_s^2}}_{\beta} + \underbrace{\frac{u^2c_s^2}{K' + Kc_s + c_s^2}}_{\gamma}\right]}$$

$$\therefore \boxed{R(u) = \frac{u}{\alpha + \beta u + \gamma u^2}}$$

(ii) Non-linear kinetics $\frac{d^2u}{d\rho^2} = \phi^2 R(u)$ , $0<\gamma<1$ , $\left.\frac{du}{d\rho}\right|_\rho = 0$ , $u(1)=0$

$$\phi^2 = \frac{\hat{r}(c_s,T)\rho_b S_g a^2}{De\, c_s} \quad ; \quad \eta = \frac{1}{\phi^2}\left.\frac{du}{d\rho}\right|_{\rho=1}$$

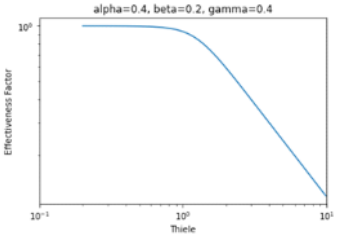
As shown in lecture, we define $v(\rho) = \frac{du}{d\rho}$ ; $F(u) = 2\int_0^u R(u)\cdot du$

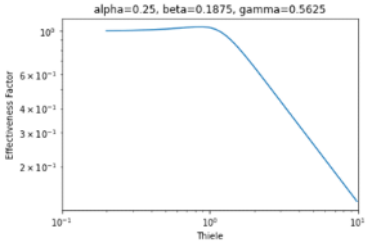
$$\phi = \int_{u_0}^{1} \frac{du}{\sqrt{F(u)-F(u_0)}} \rightarrow \text{Solve for } u_0 \text{ numerically}$$

Then use $u_0$ to solve: $\eta = \frac{v(1)}{\phi^2} = \frac{1}{\phi}\sqrt{F(1)-F(u_0)}$

Solving this system in python gives the following graphs:

(iii) (iv)




(v) Differences:

For $[\alpha = 0.25\ ;\ \beta = 0.1875\ ;\ \gamma = 0.5625]$ the effectiveness factor goes above 1. Physically this means that the diffusion rate <u>increases more</u> than avg. reaction rate (if the rate of reaction is evaluated at surface) (given the higher alpha value but lower gamma value)

## Problem 12.3

Plate catalyst pellet; First-order reaction : (w/ no external mass transfer limitations)

$$\eta = \frac{\tanh\phi}{\phi}$$

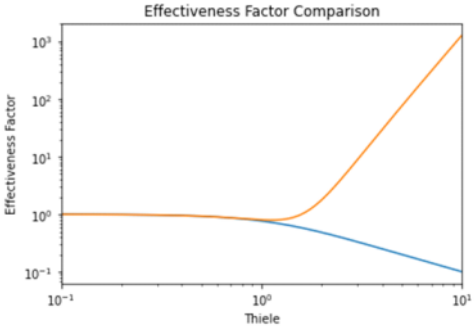
Regular perturbation theory: $\eta = 1 - \kappa_1\phi^2 + \kappa_2\phi^4 + O(\phi^6)$ ; $-\kappa_1 = -\frac{1}{V^*}\int_{V^*} \|\nabla u_1\|^2 dV^*$

$$\kappa_2 = \frac{1}{V^*}\int_{V^*} u_1^2\, dV^*$$

For plate of half-Thickness a : $\kappa_1 = \frac{1}{3}$

$\kappa_2 = \frac{2}{15}$

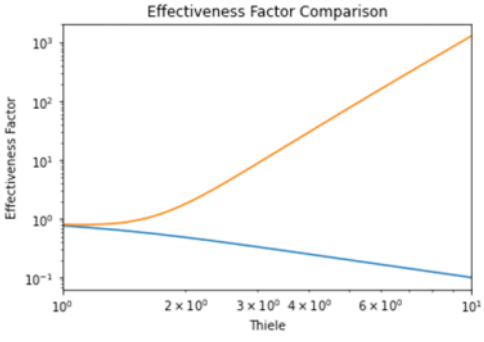
$0 < \phi < 1$




The plots below show the graphs for (i) & (ii) :

The regular perturbation is only accurate for values between $0 < \phi < 1$ as shown by the graph here;

However, the effectiveness factor approximation blows up for large $\phi$ values.




$1 < \phi < 10$

(iii) Uniform pore size distribution in plate catalyst is same as that in part (i)

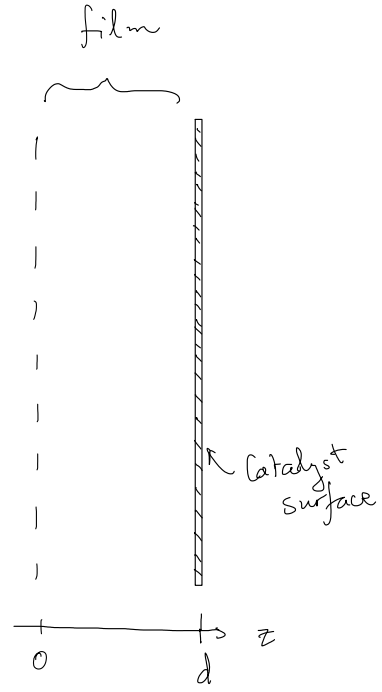
Q1] $A_1 + A_2 \rightarrow 2A_3$ ; $\hat{r}_s = \hat{k} C_{1s} C_{2s}$ (Conc. @ surface) ; ideal gas

$A_1 + A_2 - 2A_3 = 0$

$j = 1, 2, 3$

(i) 1D diffusion through a film — Bulk conc.: $C_{1f}$, $C_{2f}$

From notes: $\frac{dx_i}{dz} = \sum_{j=1}^{S} \frac{1}{cD_{ij,e}} (x_i N_j - x_j N_i)$ ; $x_i = \frac{c_i}{c_T}$ ; @ constant T and P




$i=1$ : $\frac{dx_1}{dz} = \frac{1}{cD_{11,e}}(x_1 N_1 - x_1 N_1) + \frac{1}{cD_{12,e}}(x_1 N_2 - x_2 N_1) + \frac{1}{cD_{13,e}}(x_1 N_3 - x_3 N_1)$

$\frac{dx_1}{dz} = \frac{1}{D_{11,e}}(x_1 N_1 - x_1 N_1) + \frac{1}{D_{12,e}}(x_1 N_2 - x_2 N_1) + \frac{1}{D_{13,e}}(x_1 N_3 - x_3 N_1)$

$$c \frac{dc_1}{dz} = \frac{1}{D_{12,e}}(c_1 N_2 - c_2 N_1) + \frac{1}{D_{13,e}}(c_1 N_3 - c_3 N_1)$$

$i=2$ : $\frac{dx_2}{dz} = \frac{1}{cD_{21,e}}(x_2 N_1 - x_1 N_2) + \frac{1}{cD_{22,e}}(x_2 N_2 - x_2 N_2) + \frac{1}{cD_{23,e}}(x_2 N_3 - x_3 N_2)$

Similarly,

$$c \frac{dc_2}{dz} = \frac{1}{D_{21,e}}(c_2 N_1 - c_1 N_2) + \frac{1}{D_{23,e}}(c_2 N_3 - c_3 N_2)$$

$i=3$ : $\frac{dx_3}{dz} = \frac{1}{cD_{31,e}}(x_3 N_1 - x_1 N_3) + \frac{1}{cD_{32,e}}(x_3 N_2 - x_2 N_3) + \frac{1}{cD_{33,e}}(x_3 N_3 - x_3 N_3)$

$$c \frac{dc_3}{dz} = \frac{1}{D_{31,e}}(c_3 N_1 - c_1 N_3) + \frac{1}{D_{32,e}}(c_3 N_2 - c_2 N_3)$$

(ii) Continuity eq: $\frac{dN_j}{dz} = \alpha_j r$ ; Since no reaction is happening in the film, $r = 0$

@ $z = 0$ : $\frac{dN_1}{dz} = 1r = 0$ ; $\frac{dN_2}{dz} = 1r = 0$ ; $\frac{dN_3}{dz} = -2r = 0$ } Constant flux in the film for $N_1, N_2, N_3$ ∆ conc. is equal to bulk concentration

@ $z = \delta$ : $N_1 = \alpha_1 \hat{r}$ ; $N_2 = \alpha_2 \hat{r}$ ; $N_3 = \alpha_3 \hat{r}$ ; $\hat{r}_s = \hat{k}_s C_{1s} C_{2s}$

$\alpha_1 = -1$ ; $\alpha_2 = -1$ ; $\alpha_3 = 2$

$$-N_1 = -N_2 = \frac{N_3}{2} = \hat{r}_s$$

Plugging in the (1) equation:

$$c \frac{dc_1}{dz} = \frac{1}{D_{12,e}}(-c_1(\hat{r}_s) + c_2 \hat{r}_s) + \frac{1}{D_{13,e}}(2\hat{r}_s c_1 + c_3 \hat{r}_s) = \frac{\hat{r}_s}{D_{12,e}}(-c_1 + c_2) + \frac{\hat{r}_s}{D_{13,e}}(2c_1 + c_3)$$

$$c \frac{dc_2}{dz} = \frac{1}{D_{21,e}}(-c_2 \hat{r}_s + c_1 \hat{r}_s) + \frac{1}{D_{23,e}}(2c_2 \hat{r}_s + c_3 \hat{r}_s) = \frac{\hat{r}_s}{D_{21,e}}(-c_2 + c_1) + \frac{\hat{r}_s}{D_{23,e}}(2c_2 + c_3)$$

$$c \frac{dc_3}{dz} = \frac{1}{D_{31,e}}(c_3(-\hat{r}_s) - 2\hat{r}_s c_1) + \frac{1}{D_{32,e}}(-c_3 \hat{r}_s - 2c_2 \hat{r}_s) = \frac{\hat{r}_s}{D_{31,e}}(-c_3 - 2c_1) + \frac{\hat{r}_s}{D_{32,e}}(-c_3 - 2c_2)$$

For ideal gas : $\frac{N}{V} = c = \frac{RT}{P}$ ; $\frac{dc}{dz} = \frac{d\left(\frac{RT}{P}\right)}{dz} = 0$

$a\pm$ $\quad$ $D_{31,e}$ $\quad$ $D_{32,e}$ $\quad$ $D_{33,e}$

For ideal gas : $\frac{N}{V} = c = \frac{RT}{P}$ ; $\frac{dc}{dz} = \frac{d\left(\frac{RT}{P}\right)}{dz} = 0$

Total conc. is conserved ; Total moles are conserved
(No variation in z axis)

(iii) For non-porous catalysts, Effectiveness factor is defined as actual reaction rate (including diffusion) vs the reaction rate when entire surface is at bulk conditions

$$\therefore \eta = \frac{\int_A \hat{r}_s \, dS}{A \cdot \hat{r}_f} \quad ; \quad h = \frac{\hat{r}_s}{\hat{r}_f} = \eta$$

(iv) Non-dimensionalizing: $\phi^2 = \frac{\hat{r}_f \delta}{c D_{12}}$ ; $\zeta = \frac{z}{\delta}$ ; $\hat{r}_f = \hat{k} c_{1f} c_{2f}$ ; $\hat{r}_s = h \hat{r}_f$

$$\frac{dc_1}{dz} = \frac{\hat{r}_s}{c D_{12,e}} (-c_1 + c_2) + \frac{\hat{r}_s}{c D_{13,e}} (2c_1 + c_3)$$

$$\frac{dc_1}{d(\zeta\delta)} = \frac{h\hat{r}_f}{c D_{12,e}} (-c_1 + c_2) + \frac{h\hat{r}_f}{c D_{13,e}} (2c_1 + c_3)$$

$$\frac{dc_1}{d\zeta} = \frac{h\hat{r}_f \delta}{c D_{12,e}} (-c_1 + c_2) + \frac{h\hat{r}_f \delta}{c D_{n,e}} \frac{D_{12,e}}{D_{13,e}} (2c_1 + c_3)$$

$$\boxed{\frac{dc_1}{d\zeta} = h\phi^2 (-c_1 + c_2) + h\phi^2 \frac{D_{12,e}}{D_{13,e}} (2c_1 + c_3)}$$

$$\frac{dc_2}{d(\zeta\delta)} = \frac{h\hat{r}_f}{c D_{12,e}} \frac{D_{12,e}}{D_{21,e}} (-c_2 + c_1) + \frac{h\hat{r}_f}{c D_{12,e}} \cdot \frac{D_{12,e}}{D_{23,e}} (2c_2 + c_3)$$

$$\boxed{\frac{dc_2}{d\zeta} = h\phi^2 \frac{D_{12,e}}{D_{21,e}} (-c_2 + c_1) + h\phi^2 \frac{D_{12,e}}{D_{23,e}} (2c_2 + c_3)}$$

$$\frac{dc_3}{d(\zeta\delta)} = \frac{h\hat{r}_f}{c D_{12,e}} \frac{D_{12,e}}{D_{31,e}} (-c_3 - 2c_1) + \frac{h\hat{r}_f}{c D_{13,e}} \frac{D_{12,e}}{D_{32,e}} (-c_3 - 2c_2)$$

$$\boxed{\frac{dc_3}{d\zeta} = h\phi^2 \frac{D_{12,e}}{D_{31,e}} (-c_3 - 2c_1) + h\phi^2 \frac{D_{12,e}}{D_{32,e}} (-c_3 - 2c_2)}$$

BC's:

@ $\zeta = 0$ : $c_1 = c_{1f}$ , $c_2 = c_{2f}$ (bulk concentrations)

@ $\zeta = 1$ : Diffusion rate = reaction rate

$$N_1 = -D_{1,e} \frac{dc_1}{dz} = -\hat{r}_s$$

$$N_2 = -D_{2,e} \frac{dc_2}{dz} = -\hat{r}_s$$

$$N_3 = -D_{3,e} \frac{dc_3}{dz} = 2\hat{r}_s$$

(v) Simplifying the expression using $1 = \sum_i x_i$ ; $c_i = x_i c$

$$\frac{dc_1}{d\tau_e} = h\phi^2\left(-c_1 + c_2\right) + h\phi^2 \frac{D_{12,e}}{D_{13,e}}\left(2c_1 + c_3\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left(-x_1 + x_2\right) + h\phi^2 \frac{D_{12}}{D_{13}}\left(2x_1 + x_3\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left(-x_1 + x_2\right) + h\phi^2 \frac{D_{12}}{D_{13}}\left(2x_1 + (1 - x_1 - x_2)\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left(-x_1 + x_2\right) + h\phi^2 \frac{D_{12}}{D_{13}}\left(1 + x_1 - x_2\right)$$

$$\frac{dx_1}{d\tau_e} = h\phi^2\left[x_1\left(-1 + \frac{D_{12}}{D_{13}}\right) + x_2\left(1 - \frac{D_{12}}{D_{13}}\right) + \frac{D_{12}}{D_{13}}\right]$$

Similarly, $\frac{dc_2}{d\tau_e} = h\phi^2 \frac{D_{12,e}}{D_{21,e}}\left(-c_2 + c_1\right) + h\phi^2 \frac{D_{12,e}}{D_{23,e}}\left(2c_2 + c_3\right)$ $\quad 1 - x_2 - x_1$

$$\frac{dx_2}{d\tau_e} = h\phi^2\left[\frac{D_{12}}{D_{21}}\left(-x_2 + x_1\right) + h\phi^2 \frac{D_{12}}{D_{23}}\left(1 - x_1 + x_2\right)\right]$$

$$\frac{dx_2}{d\tau_e} = h\phi^2\left[\left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right)x_1 + \left(-\frac{D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right)x_2 + \frac{D_{12}}{D_{23}}\right]$$

Combining these as an array:

$$h^{-1}\phi^{-2}\frac{d}{d\tau_e}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \left(-1 + \frac{D_{12}}{D_{13}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(-\frac{D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} D_{12}/D_{13} \\ D_{12}/D_{23} \end{bmatrix}$$

$\underline{x}$ ; $\underline{\underline{A}}$ ; $\underline{x}$ ; $\underline{b}$

(vi) Show given $\underline{x}$ is a solution;

LHS: $\underline{x}(\tau_e) = e^{n\phi^2\underline{\underline{A}}\tau_e}\,\underline{x}_f + \underbrace{n\phi^2\int_0^{\tau_e} e^{n\phi^2\underline{\underline{A}}\tau'}\,\underline{b}\,d\tau'}$

Evaluating this

$$\Rightarrow \left(n\phi^2\right)\left(n\phi^2\underline{\underline{A}}\right)^{-1} e^{n\phi^2\underline{\underline{A}}\tau'}\Big|_0^{\tau_e}\,\underline{b}$$

$$\Rightarrow \underline{\underline{A}}^{-1}\left(e^{n\phi^2\underline{\underline{A}}\tau_e} - \underline{\underline{I}}\right)\underline{b}$$

$$\underline{x}(\tau_e) = e^{n\phi^2\underline{\underline{A}}\tau_e}\,\underline{x}_f + \underline{\underline{A}}^{-1}\left(e^{n\phi^2\underline{\underline{A}}\tau_e} - \underline{\underline{I}}\right)\underline{b}$$

$$\frac{d\underline{x}}{d\tau_e} = n\phi^2\underline{\underline{A}}\,e^{n\phi^2\underline{\underline{A}}\tau_e}\,\underline{x}_f + \underline{\underline{A}}^{-1}n\phi^2\underline{\underline{A}}\,e^{n\phi^2\underline{\underline{A}}\tau_e}\,\underline{b}$$

$$= n\phi^2 e^{n\phi^2\underline{\underline{A}}\tau_e}\left(\underline{\underline{A}}\,\underline{x}_f + \underline{b}\right)$$

$h^{-1}\phi^{-2}$

ala

$$= n\phi^2 e^{n\phi^2 \underline{\underline{A}} t_a} \left( \underline{\underline{A}} \underline{x}_f + \underline{b} \right)$$

$n^{-1}\phi^{-2} \times$

$$n^{-1}\phi^{-2} \frac{d\underline{x}}{dt_a} = e^{n\phi^2 \underline{\underline{A}} t_a} \left( \underline{\underline{A}} \underline{x}_f + \underline{b} \right)$$

Solving for RHS: $\underline{\underline{A}} \underline{x} + \underline{b} \Rightarrow \underline{\underline{A}} \left( e^{n\phi^2 \underline{\underline{A}} t_a} \underline{x}_f + \underline{\underline{A}}^{-1} \left( e^{n\phi^2 \underline{\underline{A}} t_a} - \underline{\underline{I}} \right) \underline{b} \right) + \underline{b}$

$$\Rightarrow \underline{\underline{A}} e^{n\phi^2 \underline{\underline{A}} t_a} \underline{x}_f + e^{n\phi^2 \underline{\underline{A}} t_a} \underline{b} - \cancel{\underline{b}} + \cancel{\underline{b}}$$

$$\Rightarrow e^{n\phi^2 \underline{\underline{A}} t_a} \left( \underline{\underline{A}} \underline{x}_f + \underline{b} \right)$$

LHS = RHS ✓

$\underline{x}$ is a solution

(vii) Finding eigenvalues of $\underline{\underline{A}} = \begin{bmatrix} \left(-1 + \frac{D_{12}}{D_{13}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(\frac{-D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right) \end{bmatrix}$

$$\det(\underline{\underline{A}} - \lambda \underline{\underline{I}}) = \left(-1 + \frac{D_{12}}{D_{13}} - \lambda\right)\left(\frac{-D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}} - \lambda\right) - \left(1 - \frac{D_{12}}{D_{13}}\right)\left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) = 0$$

$$= \cancel{\frac{D_{12}}{D_{21}}} - \cancel{\frac{D_{12}}{D_{23}}} + \lambda - \cancel{\frac{D_{12}^2}{D_{21} D_{13}}} + \cancel{\frac{D_{12}^2}{D_{13} D_{23}}} - \lambda \frac{D_{12}}{D_{13}} + \lambda \frac{D_{12}}{D_{21}} - \lambda \frac{D_{12}}{D_{23}} + \lambda^2 - \left[ \cancel{\frac{D_{12}}{D_{21}}} - \cancel{\frac{D_{12}}{D_{23}}} - \cancel{\frac{D_{12}^2}{D_{13} D_{21}}} + \cancel{\frac{D_{12}^2}{D_{13} D_{23}}} \right]$$

$$= \lambda \left( 1 - \frac{D_{12}}{D_{13}} + \frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}} + \lambda \right) = 0$$

If $D_{21} = D_{12}$, $\boxed{\lambda = 0}$

$$1 - \frac{D_{12}}{D_{13}} + 1 - \frac{D_{12}}{D_{23}} + \lambda = 0$$

$$\boxed{\lambda = -2 + \frac{D_{12}}{D_{13}} + \frac{D_{12}}{D_{23}}}$$

$$\underline{\underline{A}} = \begin{bmatrix} \left(-1 + \frac{D_{12}}{D_{13}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(\frac{-D_{12}}{D_{21}} + \frac{D_{12}}{D_{23}}\right) \end{bmatrix}$$

Projection matrices:

$$\rightarrow \quad adj(\lambda_1 \underline{\underline{I}} - \underline{\underline{A}}) = adj(-\underline{\underline{A}}) = \begin{bmatrix} \left(+\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(+1 - \frac{D_{12}}{D_{13}}\right) \end{bmatrix}$$

$$S_1 = \frac{1}{0 - \lambda_2} adj(-\underline{\underline{A}}) = \frac{1}{\left(2 - \frac{D_{12}}{D_{13}} - \frac{D_{12}}{D_{23}}\right)} \begin{bmatrix} \left(+\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & \left(+1 - \frac{D_{12}}{D_{13}}\right) \end{bmatrix}$$

$$\rightarrow \quad (\lambda_2 \underline{\underline{I}} - \underline{\underline{A}}) = \begin{bmatrix} -2 + \cancel{\frac{D_{12}}{D_{13}}} + \frac{D_{12}}{D_{23}} + 1 - \cancel{\frac{D_{12}}{D_{13}}} & -\left(1 - \frac{D_{12}}{D_{13}}\right) \\ -\left(\frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}}\right) & -2 + \frac{D_{12}}{D_{13}} + \cancel{\frac{D_{12}}{D_{23}}} + \frac{D_{12}}{D_{21}} - \cancel{\frac{D_{12}}{D_{23}}} \end{bmatrix}$$

$$; \quad adj(\lambda_2 \underline{\underline{I}} - \underline{\underline{A}}) = \begin{bmatrix} -2 + \frac{D_{12}}{D_{13}} + \frac{D_{12}}{D_{21}} & \left(1 - \frac{D_{12}}{D_{13}}\right) \\ \frac{D_{12}}{D_{21}} - \frac{D_{12}}{D_{23}} & -1 + \frac{D_{12}}{D_{23}} \end{bmatrix}$$

$$S_2 = \frac{1}{\left(-2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{23}}\right)}\begin{bmatrix} -2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{21}} & \left(1-\frac{D_{12}}{D_{13}}\right) \\ \frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}} & -1+\frac{D_{12}}{D_{23}} \end{bmatrix}$$

$$S_1+S_2 = I \;;\quad \frac{1}{\left(2-\frac{D_{12}}{D_{13}}-\frac{D_{12}}{D_{23}}\right)}\left(\begin{bmatrix} \cancel{\frac{D_{12}}{D_{21}}}-\frac{D_{12}}{D_{13}}+2-\frac{D_{12}}{D_{13}}-\cancel{\frac{D_{12}}{D_{21}}} & \cancel{1}-\cancel{\frac{D_{12}}{D_{13}}}-\cancel{1}+\cancel{\frac{D_{12}}{D_{13}}} \\ \cancel{\frac{D_{12}}{D_{21}}}-\cancel{\frac{D_{12}}{D_{23}}}-\cancel{\frac{D_{12}}{D_{21}}}+\cancel{\frac{D_{12}}{D_{23}}} & 1-\frac{D_{12}}{D_{13}}+1-\frac{D_{12}}{D_{23}} \end{bmatrix}\right)$$

$$\Rightarrow \frac{1}{\left(2-\cancel{\frac{D_{12}}{D_{13}}}-\cancel{\frac{D_{12}}{D_{23}}}\right)}\left(\begin{bmatrix} \left(2-\cancel{\frac{D_{12}}{D_{23}}}-\frac{D_{12}}{D_{13}}\right) & 0 \\ 0 & \left(2-\cancel{\frac{D_{12}}{D_{13}}}-\frac{D_{12}}{D_{23}}\right) \end{bmatrix}\right)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \underline{\underline{I}} \quad \checkmark$$

(viii) According to Sylvester's Theorem: $f(A) = \sum_{i=1}^{k} f(\lambda_i) A_i \quad ; \quad A_i = \prod_{j=1}^{k} \frac{1}{\lambda_i - \lambda_j}(A-\lambda_j I)$

$$f(A) = f(\lambda_1)A_1 + f(\lambda_2)A_2 \qquad A_1 = \frac{1}{\lambda_1-\lambda_2}(A-\lambda_2 I) = S_1$$

$$A_2 = \frac{1}{\lambda_2-\lambda_1}(A-\lambda_1 I) = S_2$$

$$@\ t=1 : \ \underline{x}(t=1) = e^{n\phi^2 \underline{\underline{A}} t}\underline{x}_i + \underline{\underline{A}}^{-1}\left(e^{n\phi^2\underline{\underline{A}}t} - \underline{\underline{I}}\right)\underline{b}$$

Evaluating the exponentials: $f(\underline{\underline{A}}) = e^{\underline{\underline{A}}}$

$$\therefore e^{\underline{\underline{A}}} = e^{\lambda_1}A_1 + e^{\lambda_2}A_2 = e^{\lambda_1}S_1 + e^{\lambda_2}S_2$$

$$\therefore \underline{x}(1) = e^{n\phi^2}\left(e^{\lambda_1}S_1 + e^{\lambda_2}S_2\right)\underline{x}_i + \underline{\underline{A}}^{-1}\left(e^{n\phi^2}\cdot\left(e^{\lambda_1}S_1 + e^{\lambda_2}S_2\right) - \underline{\underline{I}}\right)\underline{b} \quad ; \ \lambda_1 = 0$$

$$= e^{n\phi^2}S_1\underline{x}_i + e^{n\phi^2\lambda_2}S_2\underline{x}_i + \underline{\underline{A}}^{-1}\left(e^{n\phi^2}S_1 + e^{n\phi^2\lambda_2}S_2 - \underline{\underline{I}}\right)\underline{b} \quad ; \ \underline{\underline{A}}^{-1} = \frac{1}{|A|}\,\mathrm{adj}(A)$$

$$= e^{n\phi^2}S_1\underline{x}_i + e^{n\phi^2\lambda_2}S_2\underline{x}_i + \underline{\underline{A}}^{-1}e^{n\phi^2}S_1\underline{b} + \underline{\underline{A}}^{-1}e^{n\phi^2\lambda_2}S_2\underline{b} - \underline{\underline{A}}^{-1}\underline{b}$$

$$=$$

(ix) $$\eta = \frac{\left(e_1^T x_s\right)\left(e_2^T x_s\right)}{\left(e_1^T x_f\right)\left(e_2^T x_f\right)} = \frac{\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix} x_{1s} \\ x_{2s}\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix} x_{1s} \\ x_{2s}\end{bmatrix}}{\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix} x_{1s} \\ x_{2f}\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix} x_{1f} \\ x_{2f}\end{bmatrix}} = \frac{x_{1s}\cdot x_{2s}}{x_{1f}\cdot x_{2f}}$$

$$S_1 = \frac{1}{\left(2-\frac{D_{12}}{D_{13}}-\frac{D_{12}}{D_{23}}\right)}\begin{bmatrix}\left(+\frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}}\right) & \left(1-\frac{D_{12}}{D_{13}}\right) \\ \left(\frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}}\right) & \left(+1-\frac{D_{12}}{D_{13}}\right)\end{bmatrix} = \frac{1}{\left(2-\frac{3}{2}-\frac{3}{2}\right)}\begin{bmatrix} 1-\frac{3}{2} & 1-\frac{3}{2} \\ 1-\frac{3}{2} & 1-\frac{3}{2}\end{bmatrix} = -1\begin{bmatrix} -1/2 & -1/2 \\ -1/2 & -1/2\end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2\end{bmatrix}$$

$$S_2 = \frac{1}{\left(-2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{23}}\right)}\begin{bmatrix} -2+\frac{D_{12}}{D_{13}}+\frac{D_{12}}{D_{21}} & \left(1-\frac{D_{12}}{D_{13}}\right) \\ \frac{D_{12}}{D_{21}}-\frac{D_{12}}{D_{23}} & -1+\frac{D_{12}}{D_{23}}\end{bmatrix} = \frac{1}{\left(-2+\frac{3}{2}+\frac{3}{2}\right)}\begin{bmatrix} -2+\frac{3}{2}+1 & 1-\frac{3}{2} \\ 1-\frac{3}{2} & -1+\frac{3}{2}\end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2\end{bmatrix}$$

$$\underline{b} = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix} \qquad \therefore \lambda_2 = -2 + \frac{3}{2} + \frac{3}{2} = 1$$

$$\underline{x}_s = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix} + e^{+n\phi^2} \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix} + \eta\phi^2 \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix} + \frac{1}{1} \begin{bmatrix} e^{n\phi^2} - 1 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}$$

↳ Solving for $x_{1s}$ and $x_{2s}$ from this expression to get in terms of $x_{1f}, x_{2f}$

$$\underline{x}_s = \begin{bmatrix} \frac{1}{2}x_{1f} + \frac{1}{2}x_{2f} \\ \frac{1}{2}x_{1f} + \frac{1}{2}x_{2f} \end{bmatrix} + e^{n\phi^2} \begin{bmatrix} \frac{1}{2}x_{1f} - \frac{1}{2}x_{2f} \\ -\frac{1}{2}x_{1f} + \frac{1}{2}x_{2f} \end{bmatrix} + n\phi^2 \begin{bmatrix} 6/4 \\ 6/4 \end{bmatrix}$$

$$\underline{x}_s = \begin{bmatrix} \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) x_{1f} + \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) x_{2f} \\ \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) x_{1f} + \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) x_{2f} \end{bmatrix} + n\phi^2 \begin{bmatrix} 6/4 \\ 6/4 \end{bmatrix}$$

$$\underline{x}_s = \begin{bmatrix} \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) & \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) \\ \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) & \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) \end{bmatrix} \underbrace{\begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix}}_{\underline{x}_f} + n\phi^2 \begin{bmatrix} 6/4 \\ 6/4 \end{bmatrix}$$

$$\eta = \frac{(e_1^T \underline{x}_s)(e_2^T \underline{x}_s)}{(e_1^T \underline{x}_f)(e_2^T \underline{x}_f)} = \frac{\left(\left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) x_{1f} + \left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) x_{2f} + n\phi^2 6/4\right)\left(\left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right) x_{1f} + \left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right) x_{2f} + \phi^2 n \frac{6}{4}\right)}{x_{1f} x_{2f}}$$

$$= \frac{x_{1f}^2\left(\frac{1}{4} - \frac{e^{2n\phi^2}}{4}\right) + x_{2f}^2\left(\frac{1}{4} - \frac{e^{2n\phi^2}}{2}\right) + x_{1f}x_{2f}\left(\frac{1}{2} + \frac{e^{n\phi^2}}{2}\right)^2 + x_{1f}x_{2f}\left(\frac{1}{2} - \frac{e^{n\phi^2}}{2}\right)^2 + x_{1f}\phi^2 n\frac{6}{4} + x_{2f}}{x_{1f} x_{2f}}$$

(x) Plugging in $x_{1f} = x_{2f}$

↳ will give the required expression

HW 7

Monday, 27 February 2023 3:07 PM

## Question 5.1

Shell-progressive poisoning ["shrinking core"] model:

Rate of poison deposition: $\frac{d}{dt}\left(C_{ps\infty}\,\frac{4}{3}\pi(R^3-r_c^3)\right)\rho_s = \frac{4}{3}\pi R^3 r_p \cdot \rho_s$

$C_{ps}^s$ = poison conc. at solid surface

$C_{ps}^c$ = " at core boundary

$r_c$ = radius of unpoisoned core

$C_p$ = Conc. of poison in bulk

$C_{ps\infty}$ = solid conc. of poison at saturation

$k_{g,p}$ = external interfacial MT coeff.

$k'_{sp}$ = core surface reaction rate coeff. for poison

$D_{ep}$ = effective pore diffusivity of poison

$\sigma_p$ = adsorption dist. coefficient

$S_c$ = S.A. of core / per unit volume

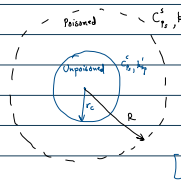
$\langle C_{ps}\rangle$ = pellet average poison conc.

$\rho_b$ = bulk density of catalyst pellet ($\frac{\text{Mass of catalyst}}{\text{Volume}}$)

$\varepsilon$ = voidage of packed bed ($\text{porosity} = \frac{\text{Volume of void space}}{\text{volume}}$)

$S_a$ = surf. area / g-cat.

$a_c$ = surf. area / volume.



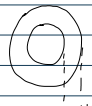

External MT limitations: Flux = $k_{gp}(C_p - C_{ps}^s)$

Rate of mass transport to the external surface = Flux · Area

$= (k_{gp}(C_p - C_{ps}^s))\,(4\pi R^2)$

$\left[\frac{mol}{s}\right] = \frac{mol}{m^2\cdot s}\cdot m^2 \cdot \frac{g}{g}\cdot$

Steady state diffusion through a spherical shell $\Rightarrow$




$W\cdot 4\pi r^2|_r - W\cdot 4\pi r^2|_{r+\Delta r} = 0$ (Diffusion within)

$\frac{d(W\cdot 4\pi r^2)}{dr} = 0$ ; $W = -D_e\frac{dC}{dr}$

$\frac{d}{dr}\left(D_e\cdot\frac{dc}{dr}\cdot r^2\right) = 0$

$D_e\frac{d^2c}{dr^2}r^2 + D_e 2r\cdot\frac{dc}{dr} = 0$

$\Rightarrow D_e\frac{d^2c}{dr^2} + \frac{2}{r}\cdot D_e\frac{dc}{dr} = 0$

general solution $\rightarrow C = \frac{c_1}{r} + c_2$

BC: 1) @ $r = r_c$, $C_{ps}^c = C$

2) @ $r = R$, $C = C_{ps}^s$

$\Rightarrow C_{ps}^c = \frac{c_1}{r_c} + c_2 \Rightarrow c_2 = C_{ps}^c - \frac{c_1}{r_c}$

$C_{ps}^s = \frac{c_1}{R} + c_2$

$(C_{ps}^s - C_{ps}^c) = c_1\left(\frac{1}{R} - \frac{1}{r_c}\right)$

$c_1 = \frac{(C_{ps}^s - C_{ps}^c)}{\left(\frac{1}{R} - \frac{1}{r_c}\right)} \Rightarrow c_2 = C_{ps}^c - \frac{1}{r_c}\frac{(C_{ps}^s - C_{ps}^c)}{\left(\frac{1}{R} - \frac{1}{r_c}\right)}$

$c_2 = C_{ps}^c - \frac{(C_{ps}^s - C_{ps}^c)}{\left(\frac{r_c}{R} - 1\right)}$

Fick's law diffusion (Flux): $\left(-D_e\frac{dc}{dr}\right)\cdot$ Area

$= +D_e\left(\frac{c_1}{r^2}\right)\cdot 4\pi r^2$

$= \frac{D_e\cdot(C_{ps}^s - C_{ps}^c)}{\left(\frac{1}{R} - \frac{1}{r_c}\right)}\cdot 4\pi$

Deposition rate at boundary: Transport of species = Reaction at the surface

$4\pi r_c^2 k_{sp} C_{ps}^c = 4\pi r_c^2 \cdot k'_{sp}\cdot C_{ps}^c$

Defining $\langle C_{ps}\rangle$ as:

$\frac{\langle C_{ps}\rangle}{C_{ps\infty}} = 1 - \left(\frac{r_c}{R}\right)^3$ ; $\xi = \frac{r_c}{R}$ $\Rightarrow \frac{d}{dt}\left(\frac{\langle C_{ps}\rangle}{C_{ps\infty}}\right) = -\frac{3}{R}r_c^2\cdot\frac{dr_c}{dt}$

$$C_{pl\infty} \cdot \frac{4}{3}\pi \frac{d}{dt}\left(R^3 - r_c^3\right) \cdot$$

$$= \rho_s C_{pl\infty} \frac{4}{3}\pi \left(-3\frac{dr_c}{dt}\right)$$

$$\frac{4}{3}\pi R^3 r_p \rho_s = 4\pi R^2 \rho_s C_{pl\infty} \frac{dr_c}{dt}$$

$$\frac{dr_c}{dt} = \frac{3R^2 r_p}{}$$

A B

1) $$\left(\frac{4}{3}\pi R^3 \rho_s\right) \cdot r_p = \left(4\pi R^2 k_{gp}\ \rho_s\right)\left(C_p - C_{ps}^s\right)$$

$$C_p = \frac{A}{B} r_p + C_{ps}^s$$

C

2) $$A r_p = \left(4\pi r_c^2 k_{sp}\right) C_{ps}^c$$

$$C_{ps}^c = \frac{A}{C} r_p$$

D

3) $$A r_p = \left(\frac{4\pi D_{ap}\ \rho_s}{\left(\frac{1}{r_c} - \frac{1}{R}\right)}\right)\left(C_{ps}^s - C_{ps}^c\right)$$

$$A r_p = D\left(C_{ps}^s - C_{ps}^c\right)$$

$$A r_p = D C_{ps}^s - \frac{D \cdot A}{C} r_p$$

$$C_{ps}^s = \left(\frac{A}{D} + \frac{A}{C}\right) r_p$$

$$\therefore \quad C_p = \frac{A}{B} r_p + \left(\frac{A}{D} + \frac{A}{C}\right) r_p$$

$$C_p = A \cdot r_p \left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)$$

$$A \cdot r_p = \frac{C_p}{\left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)}$$

4) We also know,

$$A \cdot r_p = \frac{d}{dt}\left[C_{pl\infty} \frac{4\pi}{3}\left(R^3 - r_c^3\right)\right] \cdot \rho_s$$

$$\frac{C_p}{\left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)} = \frac{4}{3}\pi \rho_s \frac{d}{dt}\left[C_{pl\infty} \cdot \left(R^3 - r_c^3\right)\right] \quad ; \quad \frac{\langle C_{pl}\rangle}{C_{pl\infty}} = \frac{R^3 - r_c^3}{R^3}$$

$$R^3 \cdot \langle C_{pl}\rangle = C_{pl\infty}\left(R^3 - r_c^3\right)$$

$$\frac{C_p}{\left(\frac{1}{B} + \frac{1}{D} + \frac{1}{C}\right)} = \frac{4}{3}\pi \rho_s \frac{d}{dt}\left[R^3 \cdot \langle C_{pl}\rangle\right]$$

$$\Rightarrow \quad \frac{3 C_p}{4\pi \rho_s R^3 \left(\frac{1}{4\pi R^2 k_{gp} \rho_s} + \frac{1}{4\pi r_c^2 k_{sp}} + \frac{1}{4\pi D_{ap} \rho_s \left(\frac{1}{r_c} - \frac{1}{R}\right)}\right)} = \frac{d}{dt}\langle C_{pl}\rangle$$

Sh term Da term $\xi$ term

$$\Rightarrow \quad \frac{3 C_p}{\rho_s \cdot D_e \cdot R^3} \left(\frac{1}{\frac{1}{R \cdot \left(\frac{2 k_{gp} R}{D_e}\right)} + \frac{1}{\left(\frac{k_{sp} \cdot R}{D_{ap}}\right)\frac{r_c^2}{R}} + \frac{1}{\rho_s\left(\frac{1}{r_c} - \frac{1}{R}\right)}}\right) = \frac{d}{dt}\langle C_{pl}\rangle$$

$$\Rightarrow \frac{3D_e C_p}{\gamma_s R^3}\left(\frac{1}{\frac{1}{R\cdot Sh_p}+\frac{1}{D_a\cdot\frac{r_c^2}{\varepsilon_c R}}+\frac{1}{\gamma_s\left(\frac{1}{r_c}-\frac{1}{R}\right)}}\right)=\frac{d}{dt}\langle C_{pR}\rangle$$

Taking $\frac{1}{R}$ as factor

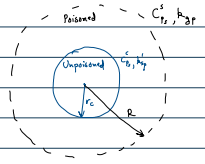
$$\Rightarrow \frac{3D_e C_p}{\gamma_s R^2\cdot\frac{1}{R}}\left(\frac{1}{\frac{1}{Sh_p}+\frac{1}{D_a\left(\frac{r_c}{R}\right)^2}+\frac{1}{\gamma_s\left(\frac{R-r_c}{R\cdot c}\right)}}\right)=\frac{d}{dt}\langle C_{lp}\rangle$$

Multiplying by tref expressions & $C_{p,\infty}$ to get:

$$\boxed{\frac{3D_e\cdot t_{ref}\,C_{p,ref}}{R^2\cdot\rho_s\cdot C_{p,\infty}}\left(\frac{1}{\frac{1}{Sh_p}+\frac{(1-\varepsilon_c)}{\varepsilon_c}+\frac{1}{D_a\varepsilon_c^2}}\right)=\frac{d}{dt'}\left[\frac{\langle C_{pR}\rangle}{C_{p,\infty}}\right]}$$

Question 2 (5-2)

+ Poisoned shell is completely inactive
+ First-order reaction (in unpoisoned core)




Diffusion in poisoned shell: $D'_{eA}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dC'_{As}}{dr}\right)=0 \qquad r_c<r<R \quad \ldots (1)$

Diffusion to pellet core is equal to the reaction rate: $D_{eA}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dC_{As}}{dr}\right)=k_{rA}\rho_s\cdot C_{As} \qquad 0<r<r_c \quad \ldots (2)$

BC's: @ $r=R$, $D'_{eA}\frac{dC'_{As}}{dr}=k_{gA}(C_A-C^s_{As})$ [External MT limitations]

@ $r=r_c$, $C_{As}=C'_{As}$ ; $D_{eA}\frac{dC_{As}}{dr}=D'_{eA}\frac{dC'_{As}}{dr}$ [Symmetry BC]

} pseudo-steady state

@ $r=0$, $C_{As}$ = finite value

Differential equations:

(1) $\frac{d}{dr}\left(r^2\frac{dC'_{As}}{dr}\right)=0$

$r^2\frac{d}{dr}C'_{As}=C_1$

$\frac{d}{dr}C'_{As}=\frac{C_1}{r^2}$

$C'_{As}=-\frac{C_1}{r}+C_2$

@ $r=r_c$:

$C'_{As}=-\frac{C_1}{r_c}+C_2$

@ $r=R$;

$\frac{dC'_{As}}{dr}=\frac{k_{gA}}{D'_{eA}}(C_A-C^s_{As})$

$\frac{C_1}{R^2}=\frac{k_{gA}}{D'_{eA}}(C_A-C^s_{As})$

$C_1=\frac{k_{gA}R^2}{D'_{eA}}(C_A-C^s_{As})$

$C_2=C'_{As}+\frac{C_1}{r_c}=C'_{As}+\frac{k_g R^2}{D'_{eA}r_c}(C_A-C^s_{As})$

$$\therefore \cancel{C_{AS}} = \frac{k_{gA} R^2}{D_{eA}} \left( -\frac{1}{r}(C_A - C_{AS}^s) + \frac{1}{r_c}(C_A - C_{AS}^s) \right) + \cancel{C_{AS}}$$

(2) $$\frac{d}{dr}\left( r^2 \frac{dC_{AS}}{dr} \right) = \frac{k_{rA} \rho_s C_{AS} \cdot r^2}{D_{eA}}$$

$$\Rightarrow r^2 \frac{d^2 C_{AS}}{dr^2} + 2r \cdot \frac{dC_{AS}}{dr} - \underbrace{\frac{k_{rA} \rho_s}{D_{eA}}}_{a} \cdot C_{AS} \cdot r^2 = 0$$

$$C_{AS} = \frac{c_1}{r} e^{-\sqrt{a} \cdot r} + \frac{c_2}{\sqrt{a} \cdot r} e^{\sqrt{a} \cdot r}$$

@ $r = R$,
$$D_e' \frac{dC_{AS}'}{dr} = k_{gA} (C_A - C_{AS}^s)$$

@ $r = r_c$,
$$C_{AS} = C_{AS}' \quad ; \quad D_e \frac{dC_{AS}}{dr} = D_e' \frac{dC_{AS}'}{dr}$$

HW 8

Monday, 6 March 2023 4:00 PM

Question 1

$A \xrightarrow{k} \text{Products}$ , $k(T) = A \exp\left(\frac{-E}{RT}\right)$ ; $c_f, T_f$ } Surface cond. ; $r = -k[A]$, $k = f(T_f)$

; No external MT resistance

; Heat transfer → only external resistance

(a) Energy balance equation: $\bar{c}_p \frac{\partial T}{\partial t} = \sum_{i}^{m} \rho_b S_{g,i} \alpha_{ij} \hat{r}_i (-\Delta H_i) + \nabla \cdot k_e \nabla T$ → Since we are assuming only external heat transfer resistance ⇒ there are no T gradients within the catalyst

Heat flux = Heat generation in the catalyst

(External HT:)

$$h(T - T_f) \cdot A = \int_V \rho_b S_g (-\Delta H) \hat{r}(c,T)\, dV = A \int_0^a \rho_b S_g (-\Delta H) \hat{r}(c,T)\, dx$$

Mass balance equation: $\varepsilon \frac{\partial c_j}{\partial t} = \sum_{k=1}^{5} \nabla \cdot (D_{jk,e} \nabla c_k) + \sum_{i}^{m} \rho_b S_{g,i} \alpha_{ij} \hat{r}_i$

Assumptions: 1) Catalyst surface is "omnipresent" in the pellet → i.e. reaction occurs everywhere in the medium

Rewriting the differential equations in non-dimensional form (for given single reaction):

Conc. PDE: $\varepsilon \frac{\partial c_j}{\partial t} = \sum_{k=1}^{5} \nabla \cdot (D_{jk,e} \nabla c_k) + \sum_{i}^{m} \rho_b S_{g,i} \alpha_{ij} \hat{r}_i$ $\quad \alpha_{ij} = -1$

$$\varepsilon \frac{\partial c}{\partial t}^{\to 0} = \nabla \cdot D_e \nabla c - \rho_b S_g \hat{r}(c,T) \quad @\ SS$$

$$0 = \nabla \cdot D_e \nabla (u \cdot c_f) - \rho_b S_g R(u,v) \cdot \hat{r}(c_f, T_f)$$

$D_e$ = constant

$$0 = c_f (\nabla \cdot D_e \nabla u) - \rho_b S_g R(u,v) \hat{r}(c_f, T_f)$$

$$0 = \frac{D_e \cancel{c_f} \nabla^2 u}{\cancel{c_f}} - \rho_b S_g \frac{1}{\cancel{c_f}} k(T_f) \cancel{c_f} R(u,v)$$

$$0 = \nabla^2 u - \frac{\rho_b S_g k(T_f)}{D_e} \cdot R(u,v) \quad \Rightarrow \text{1D plate pellet}$$

$$\nabla^2 u = \frac{\rho_b S_g k(T_f) R(u,v)}{D_e}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\rho_b S_g k(T_f) R(u,v)}{D_e} \quad ; \quad x = \rho \cdot a$$

$$\frac{\partial^2 u}{a^2 \partial \rho^2} = \frac{\rho_b S_g k(T_f)}{D_e} \cdot R(u,v)$$

$$\frac{\partial^2 u}{\partial \rho^2} = \frac{\rho_b S_g k(T_f) a^2}{} \cdot R(u,v) \quad ; \quad R(u,v) = \frac{\hat{r}(c,T)}{} = \frac{k(T) \cdot c}{} = u \cdot \frac{A \exp\left(\frac{E}{RT}\right)}{} = u \cdot \exp\left(\frac{-E}{} \left( \frac{}{T_f} - 1 \right)\right)$$

$u \quad dp \qquad\qquad \nu e$

$$\frac{\partial^2 u}{\partial p^2} = \underbrace{\frac{\rho_b S_g k(T_f) a^2}{D_e}}_{\phi^2} \cdot R(u,\nu) \quad ; \quad R(u,\nu) = \frac{\hat{r}(c,T)}{\hat{r}(c_f,T_f)} = \frac{k(T)\cdot c}{k(T_f)\cdot c_f} = u \cdot \frac{A \exp\left(\frac{E}{RT}\right)}{A \exp\left(\frac{-E}{RT_f}\right)} = u \cdot \exp\left(\frac{-E}{RT_f}\left(\frac{T_f}{T} - 1\right)\right)$$

$$R(u,\nu) = u \cdot \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right)$$

Boundary conditions:

1) @ $p = 1$, $u = 1$

2) @ $p = 0$, $\frac{\partial u}{\partial x} = 0$ (Bounded conc.)

$$\therefore \quad \frac{\partial^2 u}{\partial p^2} = \phi^2 \cdot u \cdot \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right) = \cdot u$$

$$u = c_1 e^{\sqrt{A}p} + c_2 e^{-\sqrt{A}p} \quad \Rightarrow \quad 1 = c_1 e^{\sqrt{A}} + c_2 e^{-\sqrt{A}}$$

$$\frac{\partial u}{\partial p} = c_1\sqrt{A}\, e^{\sqrt{A}p} + c_2(-\sqrt{A})\, e^{-\sqrt{A}p}$$

$$0 = c_1\sqrt{A} - c_2\sqrt{A}$$

$$c_1 = c_2$$

$$1 = c_1 e^{\sqrt{A}} + c_1 e^{-\sqrt{A}}$$

$$c_1 = \frac{1}{e^{\sqrt{A}} + e^{-\sqrt{A}}} = \frac{1}{2\cosh(\sqrt{A})}$$

$$\therefore \quad u = c_1\left(e^{\sqrt{A}p} + e^{-\sqrt{A}}\right)$$

$$= \frac{1}{2\cosh(\sqrt{A})}\left(2\cdot\cosh(\sqrt{A}\cdot p)\right)$$

$$\boxed{u = \frac{\cosh(\sqrt{A}\cdot p)}{\cosh(\sqrt{A})}} \quad ; \quad \sqrt{A} = \sqrt{\phi^2 \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right)} = \Psi$$

Solving for temperature:

$$h(T - T_f) = \int_0^a \rho_b S_g (-\Delta H)\, \hat{r}(c,T)\cdot dx \quad ; \quad \hat{r}(c,T) = R(u,\nu)\cdot \hat{r}(c_f,T_f)$$

$$= u \cdot \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right) \cdot k(T_f)\cdot c_f$$

$$h(\nu\cdot T_f - T_f) = \int_0^1 \rho_b S_g (-\Delta H)\, k(T_f) c_f \cdot u \cdot \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right) \cdot d(p\cdot a)$$

$$(\nu - 1) = \int_0^1 u \underbrace{\frac{\rho_b \cdot S_g k(T_f) a^2}{D_e}}_{\phi^2} \cdot \underbrace{\frac{c_f \cdot D_e (-\Delta H)}{h T_f \cdot a}}_{\beta} \cdot \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right) \cdot dp \quad \Rightarrow \quad (\nu - 1) = \phi^2 \beta \int_0^1 R(u,\nu)\cdot dp$$

$$(\nu - 1) = \phi^2 \beta \int_0^1 \underbrace{\exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right)}\, u \cdot dp$$

$$(\nu - 1) = \phi^2 \beta \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right) \int_0^1 u \cdot dp$$

$$(\nu - 1) = \phi^2 \beta \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right) \int_0^1 u \cdot dp \quad ; \quad \Psi = \phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)$$

$$\Psi^2 = \phi^2 \exp\left(\gamma\left(1 - \frac{1}{\nu}\right)\right)$$

$$(\nu - 1) = \beta \Psi^2 \int_0^1 u \cdot dp$$

$$(\nu - 1) = \beta \Psi^2 \int_0^1 \frac{\cosh(\Psi p)}{\cosh(\Psi)} \cdot dp$$

$$(\nu - 1) = \frac{\beta \Psi^2}{\cosh(\Psi)} \frac{\sinh(\Psi)}{\Psi}$$

$$\boxed{(\nu - 1) = \beta \Psi \tanh(\Psi)} \quad ; \quad \Psi = \phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)$$

$$\eta = \frac{r_{obs}}{\hat{r}_s} = \frac{\int_V r \, dV}{r_s V} = \frac{\int_V \frac{r}{r_s} \cdot A \, da}{A \cdot a} = \int_0^1 R(u,v) \cdot dp \quad ;$$

$$\boxed{\eta = \frac{(\nu - 1)}{\phi^2 \beta}}$$

(b) Limiting case:

+ Small $\phi$: $\phi \to 0$ ; $(\nu - 1) = \beta \left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)$ ($\phi \to 0$)

$\nu - 1 = 0 \Rightarrow \boxed{\nu \to 1}$

$$\eta = \frac{\Psi \tanh(\Psi)}{\phi^2} = \frac{\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)}{\phi^2} \approx \frac{\tanh(\phi)}{\phi} \approx 1 \; ; \quad \boxed{\eta \to 1}$$

+ Large $\phi$: $\phi \to \infty$ ; $\boxed{\eta \to 0}$

$(\nu - 1) = \beta \left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)$ ($\tanh \to 1$)

$$\frac{(\nu - 1)}{\exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)} = \beta \phi$$

$\boxed{\nu \to 1}$

(C) $\eta = f(\nu, \beta, \phi)$ ; $\nu = f(\Psi, \phi)$

$(\nu - 1) = \beta \Psi \tanh \Psi$

$(\nu - 1) = \beta \phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)$

↓

Solving for $\nu$ for given $\phi$ values → plug in $\eta$ & plot $\eta$ vs. $\phi$

HW 9

Monday, 20 March 2023 4:00 PM

## Question 1

Steady state, non-isothermal catalyst pellet ; First order reaction

$$\nabla^2 u = \phi^2 u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\} \qquad 0<\rho<1 \qquad ; \quad \nabla^2 = \frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{d}{d\rho}\right)$$

(i) $u_N = 1 + (1-\rho^2)\sum_{n=0}^{N-1} \alpha_n^{(N)} P_n(\rho^2)$

$$u_0 = 1 + (1-\rho^2)\sum_{n=0}^{0} \alpha_n^{(N)} P_n(\rho^2) = 1 + (1-\rho^2)\left(\alpha_0 P_0(\rho^2)\right) \qquad ; \quad P_0(\rho^2) = (-1)^0 \frac{0!}{(0!)}\frac{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+0\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+0\right)}\frac{\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{1}{2}+0\right)}\rho^{2(0)}$$

$$P_0(\rho^2) = 1$$

$$\therefore u_1 = 1 + (1-\rho^2)(\alpha_0) = 1 + \alpha_0 - \alpha_0\rho^2$$

$$P_1(\rho^2) = \sum_{k=0}^{1} (-1)^k \frac{(n!)}{(n-k)!}\frac{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+n+k\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+n\right)}\frac{\Gamma\left(\frac{1}{2}q+\frac{1}{2}+k\right)}{\Gamma\left(\frac{1}{2}q+\frac{1}{2}+k\right)}\rho^{2k}$$

$$= 1\cdot 1\cdot \frac{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+1\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+1\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}+0\right)}\rho^0 + \frac{(-1)(1)\ \Gamma\left(\frac{1}{2}q+\frac{3}{2}+2\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+1\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}+1\right)}\rho^2$$

$$= 1 - \frac{\Gamma\left(\frac{q}{2}+\frac{7}{2}\right)\Gamma\left(\frac{q}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{q}{2}+\frac{5}{2}\right)\Gamma\left(\frac{q}{2}+\frac{3}{2}\right)}\rho^2$$

We know, $\Gamma(n) = (n-1)!$

$$\therefore \Gamma\left(\frac{q+7}{2}\right) = \left(\frac{q+7}{2}-1\right)! \quad \Rightarrow \quad \left(\frac{q+5}{2}\right)! = \frac{(q+5)!}{2!}$$

$$\Gamma\left(\frac{q+1}{2}\right) = \frac{(q-1)!}{2!}$$

$$\Gamma\left(\frac{q+5}{2}\right) = \frac{(q+3)!}{2!}$$

$$\Gamma\left(\frac{q+3}{2}\right) = \frac{(q+1)!}{2!}$$

$$P_1(\rho^2) = 1 - \left[\frac{\frac{(q+5)!}{2}\cdot\frac{(q-1)!}{2}!}{\frac{(q+3)!}{2}!\ \frac{(q+1)!}{2}!}\right]\rho^2$$

$$\frac{q+5}{2}-1 = \frac{q+3}{2}$$

$$\frac{q-1}{2}-1 = \frac{q-3}{2}$$

$$\frac{q+1}{2}-1 = q-1$$

$$P_1(\rho^2) = 1 - \frac{\frac{(q+5)}{2}\frac{(q+3)}{2}!\ \frac{(q-1)}{2}!}{\frac{(q+3)}{2}!\ \frac{(q+1)}{2}\frac{(q-1)}{2}!}\rho^2$$

$$P_1(\rho^2) = 1 - \frac{q+5}{(q+1)}\rho^2$$

From the question: collocation for N-point is the root of $(N+1)^{th}$ basis function

Root of: $P_1(\rho^2) = 1 - \frac{(q+5)}{(q+1)}\rho^2 = 0$

$$\rho^2 = \frac{(q+1)}{(q+5)}$$

$$\boxed{\rho^* = \left[\frac{(q+1)}{(q+5)}\right]^{1/2}}$$

(ii) $\eta = \frac{(q+1)}{\phi^2}\left.\frac{du}{d\rho}\right|_{\rho=1}$ ; $\frac{du_1}{d\rho} = -2\alpha_0\rho$

$$\left.\frac{du_1}{d\rho}\right|_{\rho=1} = -2\alpha_0$$

Solving for $\alpha_0$: $u_1 = 1 + (1-\rho^2)\alpha_0$ @ $\rho = \rho^* \Rightarrow u_1 = U$

$$(U-1) = (1-\rho^{*2})\alpha_0$$

$$\alpha_0 = \frac{(U-1)}{(1-\rho^{*2})} = \frac{(U-1)}{\left[1 - \frac{(q+1)}{(q+5)}\right]} = \frac{(U-1)(q+5)}{4}$$

$$\therefore \quad \eta = \frac{(q+1)}{\phi^2}(-2\alpha_0)$$

$$= \frac{(q+1)}{\phi^2}\frac{(-2)(U-1)(q+5)}{4}$$

$$\boxed{\eta = \frac{(q+1)(q+5)(1-U)}{2\phi^2}}$$

(iii) From the transport equation: $\nabla^2 u = \phi^2 u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}$

$$\nabla^2 u_1 = \nabla^2(1 + (1-\rho^2)\alpha_0)$$

$$= \frac{1}{\rho^q}\frac{d}{d\rho}\left(\rho^q \frac{d}{d\rho}[1 + (1-\rho^2)\alpha_0]\right)$$

$$= \frac{1}{\rho^q}\frac{d}{d\rho}\left(\rho^q \cdot (-2\rho)\alpha_0\right) = \frac{1}{\rho^q}\frac{d}{d\rho}\left(-2\rho^{(q+1)}\alpha_0\right)$$

$$= \frac{1}{\rho^q}(-2\alpha_0)(q+1)\rho^q$$

$$= -2\frac{(U-1)(q+5)(q+1)}{4} = \frac{(1-U)(q+5)(q+1)}{2}$$

@ $\rho = \rho^*$, $u_1 = u = U$

$$\therefore \nabla^2 u_1 = \phi^2 U \exp\left\{\frac{\gamma\beta(1-U)}{1+\beta(1-U)}\right\}$$

$$\boxed{\frac{(1-U)(q+5)(q+1)}{2} = \phi^2 U \exp\left\{\frac{\gamma\beta(1-U)}{1+\beta(1-U)}\right\}}$$

(iv) $q = 1$

$$(1-U)(6) = \phi^2 U \exp\left\{\frac{\gamma\beta(1-U)}{1+\beta(1-U)}\right\} \Rightarrow U \text{ as } f(\gamma, \beta, \phi^2)$$

$\eta$ @ $q = 1$:

$$\eta = \frac{(q+1)(q+5)(1-U)}{2\phi^2} = \frac{6(1-U)}{\phi^2}$$

$$\eta\phi^2 = 6(1-U)$$

$$\frac{d(\eta\phi^2)}{d\phi} = \frac{d}{d\phi}(6-6U) = -6\frac{dU}{d\phi}$$

Using python, for a given $\gamma, \beta$ value we can solve for $\eta$ as a function of $\phi$

↓

This can be plotted for various initial guesses for implicit function of $U$

↓

$\underline{\phi}$ & $\bar{\phi}$ represent the finite values for which multiplicity may appear

↓

Numerically it is shown in the graph

→ See python

Numerically it is shown in the graph

$\underline{\phi} = 0.3$ ; $\bar{\phi} = 0.6$ for $\gamma = 4$, $\beta = 10$ } See python file attached

Solving for the criteria: $\nabla^2 u = \phi^2 u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}$

If solution is not unique: $\underline{u} \le u \le \bar{u}$ everywhere in $\Omega$

By letting $f(u) = u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}$

$\nabla^2 \bar{u} = -\phi^2 f(\bar{u})$ in $\Omega$, $\bar{u} = 1$ on $\partial\Omega$

$\nabla^2 \underline{u} = -\phi^2 f(\underline{u})$ in $\Omega$, $\underline{u} = 1$ on $\partial\Omega$

$$\int_\Omega \left[(\underline{u}-1)\nabla^2\bar{u} - (\bar{u}-1)\nabla^2\underline{u}\right] dV = -\phi^2 \int_\Omega \left[(\underline{u}-1) f(\bar{u}) - (\bar{u}-1) f(\underline{u})\right] dV$$

$$F(u) = \frac{f(u)}{(u-1)}$$

$$\Rightarrow -\phi^2 \int_\Omega (\underline{u}-1)(\bar{u}-1)F(\bar{u}) - (\bar{u}-1)(\underline{u}-1)F(\underline{u})\ dV$$

$$\Rightarrow \boxed{-\phi^2 \int_\Omega \left[(\underline{u}-1)(\bar{u}-1)\right]\left(F(\bar{u}) - F(\underline{u})\right) dV = 0}$$

Sufficient condition for uniqueness regardless of $\phi^2$ value is $F(u)$ has to monotonic in range $0 \le u \le 1$ ; Additionally, this equation is a requirement for multiplicity

$\therefore \frac{dF}{du} < 0$, $0 \le u \le 1$

Differentiating, we get: $\frac{dF}{du} = \frac{d}{du}\left(\frac{f(u)}{(u-1)}\right) = \frac{f'(u)(u-1) - f(u)}{(u-1)^2}$ ; $f(u) = u \cdot \exp\left\{\frac{\beta\gamma(1-u)}{1+\beta(1-u)}\right\}$ } Wolfram Alpha

$$\Rightarrow \frac{\exp\left\{\frac{\beta\gamma(1-u)}{1+\beta(1-u)}\right\}\left(\frac{-(\gamma+2)\beta u + \beta^2(u-1)^2 + 2\beta + 1}{(1+\beta(u-1))^2}\right)\cdot(u-1) - u\cdot\exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}}{(u-1)^2} < 0$$

$$\Rightarrow \frac{\left[-(\gamma+2)\beta u + \beta^2(u-1)^2 + 2\beta + 1\right](u-1)}{(1-\beta(u-1))^2} - u < 0$$

$$\Rightarrow \left[-(\gamma+2)\beta u + \beta^2(u-1)^2 + 2\beta + 1\right](u-1) - u\cdot(1-\beta(u-1))^2 < 0$$

$$\left[-(\gamma+2)\beta u + \beta^2(u^2+1-2u) + 2\beta + 1\right](u-1) - u\left(1 + \beta^2(u-1)^2 - 2\beta(u-1)\right) < 0$$

$$\Rightarrow \left[-\beta\gamma u - 2\beta u + u^2\beta^2 + \beta^2 - 2u\beta^2 + 2\beta + 1\right](u-1) - \left(u + u\beta^2(u-1)^2 - 2\beta u(u-1)\right) < 0$$

$$\left[-\beta\gamma u^2 - 2\beta u^2 + u^3\beta^2 + u\beta^2 - 2u^2\beta^2 + 2u\beta + u + \beta\gamma u + 2\beta u - u^2\beta^2 - \beta^2 + 2u\beta^2 - 2\beta - 1 - u - u^3\beta^2 - u\beta^2 + 2u^2\beta^2 - 2\beta u^2 + 2\beta u\right] < 0$$

↓

This can be solved to get condition for inequality relating $\beta, \gamma, u$ using the roots of cubic to less than 0

HW 11

Monday, 27 March 2023 8:36 PM

Question 1

CSTR ; $B + \sigma S \rightarrow 2B + \dots$ ; $K_s$ = michaelis constant

($B$: Biomass; $S$: mass of substrate)

$\mu$ = max. specific growth

(i) Steady states ? Desirable steady states ? Inequality involving the operating parameter, $\theta$ and $s_f$ ?

Assuming perfectly-mixed : CSTR design equation $\frac{d(V c_j)}{dt} = q_{in} c_{jf} - q_{out} c_j + V \sum_{i=1}^{m} \alpha_{ij} r_i$

Constant V → $V \frac{dc_j}{dt} = q_{in} c_{jf} - q_{out} c_j + V r_B$

Mass equation (for B):

$$V \frac{db}{dt} = -q_{out} b + V r_B \quad (\text{No } b \text{ coming})$$

$$V \frac{db}{dt} = -q_{out} b + V \left( \frac{\mu_s b}{K_s + s} \right)$$

$b = \frac{u s_f}{\sigma}$ ; $s = v \cdot s_f$ ; $t = \frac{\tau}{\mu}$ ; $K_s = \kappa \cdot s_f$

$$\Rightarrow V \frac{d\left(\frac{u s_f}{\sigma}\right)}{d\left(\frac{\tau}{\mu}\right)} = -q_{out} \left(\frac{u s_f}{\sigma}\right) + V \left( \frac{\mu \cdot v s_f \cdot \frac{u s_f}{\sigma}}{\kappa s_f + v s_f} \right)$$

$$\frac{s_f V \mu}{\sigma} \cdot \frac{du}{d\tau} = -q_{out} \cdot \frac{u s_f}{\sigma} + \frac{V}{\sigma} \left( \frac{\mu v \cdot u}{\kappa + v} \right) s_f$$

$$(V\mu) \cdot \frac{du}{d\tau} = -q_{out} \cdot u + V \left( \frac{\mu v \cdot u}{\kappa + v} \right)$$

$$V\mu \frac{du}{d\tau} = -\frac{V}{\theta} \cdot u + V \left( \frac{\mu v \cdot u}{\kappa + v} \right)$$

$$\Rightarrow \boxed{\frac{du}{d\tau} = -\frac{1}{(\mu\theta)} u + \left( \frac{v \cdot u}{\kappa + v} \right)}$$

$q = \frac{V}{\theta}$ → Total volume; → average residence time

Mass equation for C : $V \frac{ds}{dt} = q_{in} s_f - q_{out} s + V r_s$ ; $r_B = \frac{-r_s}{\sigma}$

$$V \frac{d(v \cdot s_f)}{d\left(\frac{\tau}{\mu}\right)} = q_{in} s_f - q_{out} (v \cdot s_f) - \frac{V}{\sigma} \left( \frac{\mu v \cdot u}{\kappa + v} \right) s_f \cdot \sigma$$

$$\mu V s_f \frac{dv}{d\tau} = s_f \left( q_{in} - q_{out} v \right) - V \left( \frac{\mu v \cdot u}{\kappa + v} \right) s_f$$

For CSTR : $q_{in} = q_{out} = \frac{V}{\theta}$

$$\therefore \mu V \frac{dv}{d\tau} = \frac{V}{\theta} (1 - v) - V \left( \frac{\mu v \cdot u}{\kappa + v} \right)$$

$$\boxed{\frac{dv}{d\tau} = \frac{1}{(\mu\theta)} (1 - v) - \left( \frac{v \cdot u}{\kappa + v} \right)}$$

$q = \frac{V}{\theta}$ → mean

$(\kappa + v) \sim v$.

@ SS $\frac{dv}{d\tau} = \frac{du}{d\tau} = 0 = F(u, v)$

Condition I : $v = 1$ , $u = 0$ → No biomass formation, just substrate feed going in and out unreacted

↳ Undesirable

Condition II :

$$\frac{du}{d\tau} = -\frac{u}{\mu\theta} + \frac{v \cdot u}{\kappa + v} = 0$$

Solving for v :

$$\frac{v}{\kappa + v} = \frac{1}{\mu\theta}$$

$$v(\mu\theta) = \kappa + v$$

Energy balance : $\frac{dT}{d\tau} = -\eta - \beta [\eta - \eta_c] + B Da R(\xi, \eta)$

Assuming adiabatic reactor : $\beta = 0$

$$\frac{d\eta}{d\tau} = -\eta + B Da R\left(\frac{\eta}{B}, \eta\right) \quad ; \quad B\xi = \eta$$

Specific steady state:

$$\eta = B Da R\left(\frac{\eta}{B}, \eta\right) \quad ; \quad R(\xi, \eta) =$$

CSTR Adiabatic energy balance : $T - T_f = \frac{(-\Delta H_R) X}{\sum \theta_i c_{i_f}}$ ;

$$\frac{r(c,T)}{r(c_i,T_j)} = \frac{r_a}{r_a^b} = \frac{\frac{\chi_a k}{K_a + c}}{\frac{\chi_a c_j k_j}{k_j^2 + c_j}} = \frac{sb}{(K_a + s)} \cdot \frac{(k_j^2 + c_j)}{c_j k_j}$$

$$X = \text{conversion} = \frac{N_A^0 - N_j}{\alpha_j}$$

Solving for $v$: $\quad \frac{v}{\kappa+v} = \frac{1}{\mu\theta}$

$v(\mu\theta) = \kappa + v$

$v(\mu\theta - 1) = \kappa$

$v = \frac{\kappa}{\mu\theta - 1}$ ) Plug in $v$

$\frac{(1-v)}{\mu\theta} + \frac{v-u}{\kappa+v} = 0$

Solving for $u$:

$\Rightarrow \quad \frac{v-u}{\kappa+v} = \frac{v-1}{\mu\theta}$

$u = \frac{(v-1)(\kappa+v)}{v\mu\theta}$

$u = \frac{\left(\frac{\kappa}{(\mu\theta-1)} - 1\right)\left(\kappa + \frac{\kappa}{(\mu\theta-1)}\right)}{v\mu\theta}$

$u = \frac{(\kappa - (\mu\theta-1))(\kappa(\mu\theta-1)+\kappa)}{(\mu\theta-1)v\mu\theta}$

$u = \frac{(\kappa - (1-\mu\theta))(1-\mu\theta)(-\kappa)}{(\mu\theta-1)(v\mu\theta)}$

$u = \frac{(\kappa+1) - \theta\mu}{(1-\theta\mu)}$

Condition II:

$(u^*, v^*) = \left[\frac{(\kappa+1)-\theta\mu}{(1-\theta\mu)}, \frac{\kappa}{(\mu\theta-1)}\right]$

↳ This state is desirable

Inequality for operation: $u > 0$

$\frac{(\kappa+1)-\theta\mu}{(1-\theta\mu)} > 0$

(b) Evaluating stability. $F = \frac{du}{d\tau}$, $\frac{dv}{d\tau} = G$ vars $(u, v)$

Jacobian matrix:

$J = \begin{pmatrix} \frac{\partial F}{\partial u} & \frac{\partial F}{\partial v} \\ \frac{\partial G}{\partial u} & \frac{\partial G}{\partial v} \end{pmatrix}_{(u^*, v^*)}$

$\frac{\partial F}{\partial u} = -\frac{1}{\mu\theta} + \frac{v}{(\kappa+v)}$ $\quad \frac{\partial G}{\partial u} = \frac{-v}{\kappa+v}$

$\frac{\partial F}{\partial v} = \frac{(\kappa+v)u - vu}{(\kappa+v)^2}$ $\quad \frac{\partial G}{\partial v} = -\frac{1}{\mu\theta} - \left(\frac{(\kappa+v)u - vu}{(\kappa+v)^2}\right)$

Plugging in the values of steady state we can obtain determinant ($\Delta$) and trace ($\tau$):

For stability: $\tau^2 - 4\Delta < 0$, $\Delta \neq 0$ and $\tau^2 \neq 4\Delta$ & $\Delta < 0$

$\Delta = \det(J)$

$J = \begin{pmatrix} -\frac{1}{\mu\theta} + \frac{v}{(\kappa+v)} & \frac{\kappa}{(\kappa+v)^2} \\ \frac{-v}{(\kappa+v)} & -\frac{1}{\mu\theta} - \left(\frac{\kappa u}{(\kappa+v)^2}\right) \end{pmatrix}$

Condition 1: $v = 1$, $u = 0$

Condition 2: $v = \frac{\kappa}{(\mu\theta-1)}$, $u = \frac{(\kappa+1)-\theta\mu}{(1-\theta\mu)}$

$1+u = \frac{1-\theta\mu+(\kappa+1)-\theta\mu}{(1-\theta\mu)} = \frac{2+\kappa-2\theta\mu}{(1-\theta\mu)}$

$\Rightarrow \frac{\kappa}{(\mu\theta-1)} \cdot \frac{1}{\kappa + \frac{\kappa}{(\mu\theta-1)}} = \frac{\frac{\kappa}{(\mu\theta-1)}}{\frac{\kappa(\mu\theta-1)+\kappa}{(\mu\theta-1)}} = \frac{\kappa}{\kappa(\mu\theta-1)+\kappa}$

$= \frac{1}{(\mu\theta-1)+1}$

$= \frac{1}{\mu\theta}$

$J_1 = \begin{pmatrix} -\frac{1}{\mu\theta} + \frac{1}{(\kappa+1)} & \frac{\kappa}{(\kappa+1)^2} \\ \frac{-1}{(\kappa+1)} & -\frac{1}{\mu\theta} \end{pmatrix}$

$J_2 = \begin{pmatrix} 0 & \frac{1}{(\mu\theta)^2} \\ -\frac{1}{\mu\theta} & -\frac{1}{\mu\theta}\left(1 + \frac{1}{\mu\theta}\right) \end{pmatrix}$

$\Rightarrow \frac{\kappa}{\quad} = \frac{1}{\quad}$

$v = \frac{\mu\theta'}{\sigma}$

$$J_2 = \begin{pmatrix} 0 & \frac{1}{(\mu\theta)^2} \\ -\frac{1}{\mu\theta} & -\frac{1}{\mu\theta}\left(1+\frac{4}{(\mu\theta-1)}\right) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{1}{(\mu\theta)^2} \\ -\frac{1}{\mu\theta} & -\frac{1}{\mu\theta}\left(\frac{2+\kappa-2\theta\mu}{1-\theta\mu}\right) \end{pmatrix}$$

$$\det(J_1) = -\frac{1}{\mu\theta}\left(-\frac{1}{\mu\theta}+\frac{1}{(\kappa+1)}\right)+\frac{\kappa}{(\kappa+1)^3}$$

$$= \frac{1}{(\mu\theta)^2} - \frac{1}{\mu\theta(\kappa+1)} + \frac{\kappa}{(\kappa+1)^3}$$

$$\det(J_2) = \frac{1}{(\mu\theta)^2} > 0$$

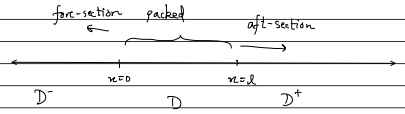
$$\tau_{J_1} = -\frac{2}{\mu\theta}+\frac{1}{(\kappa+1)}$$

$$\tau_{J_2} = -\frac{1}{\mu\theta}\left(\frac{2+\kappa-2\theta\mu}{1-\mu\theta}\right)$$

(iii) Oscillations can exist when there are complex values for eigenvalues for the Jacobian.

Question 2

PBR, axial dispersion model, No reaction; Tracer experiment




(i) From class notes:

(ii) each section has a general equation: $D^- \frac{\partial^2 c^-}{\partial x^2} - v_x \frac{\partial c^-}{\partial x} = \frac{\partial c^-}{\partial t}$ ; $c^- = \frac{u^- \cdot Q}{Al}$ ; $\tau = \frac{t \cdot Al}{q} = t$

$x = \xi \cdot l$ ; $v_x = \frac{Pe^- \cdot D^-}{l}$

$$\Rightarrow D^- \frac{\partial^2\left(\frac{u^-\theta}{Al}\right)}{\partial(\xi l)^2} - \frac{Pe^- D^-}{l}\frac{\partial\left(\frac{u^-\theta}{Al}\right)}{\partial(\xi l)} = \frac{\partial\left(\frac{u^-\theta}{Al}\right)}{\partial\left(\frac{\tau \cdot Al}{q}\right)}$$

$$\frac{D^-\theta}{Al}\cdot\frac{1}{l^2}\frac{\partial^2 u^-}{\partial\xi^2} - \frac{Pe^- D^-}{l}\cdot\frac{\theta}{Al}\cdot\frac{1}{l}\frac{\partial u^-}{\partial\xi} = \frac{\theta}{Al}\frac{q}{Al}\frac{\partial u^-}{\partial\tau}$$

$$\Rightarrow \frac{D^-\theta}{Al^3}\frac{\partial^2 u^-}{\partial\xi^2} - \frac{Pe^- D^-\theta}{Al^3}\frac{\partial u^-}{\partial\xi} = \frac{\theta q l}{Al^2 A}\frac{\partial u^-}{\partial\tau}$$

$$\Rightarrow D\frac{\partial^2 u^-}{\partial\xi^2} - Pe^- D^-\frac{\partial u^-}{\partial\xi} = \frac{lq}{A}\frac{\partial u^-}{\partial\tau}$$

$$\boxed{\frac{\partial^2 u^-}{\partial\xi^2} - Pe^-\frac{\partial u^-}{\partial\xi} = Pe^-\frac{\partial u^-}{\partial\tau}}$$

$Pe = \frac{lv}{D} = \frac{l\cdot q}{D\cdot A}$

Similarly

$$\boxed{\frac{\partial^2 u^+}{\partial\xi^2} - Pe^+\frac{\partial u^+}{\partial\xi} = Pe^+\frac{\partial u^+}{\partial\tau}}$$

$$\boxed{\frac{\partial^2 u}{\partial\xi^2} - Pe\frac{\partial u}{\partial\xi} = Pe\frac{\partial u}{\partial\tau}}$$

Boundary conditions: @ $t=0$ and $x=0$, $c^- = \frac{Q\delta(t)+c}{Al}$ — 2 BC & 1 IC

@ $\tau=0$, $\xi=0$, $u=u^- = \delta(\tau)$

$x=l$, $c=c^+$

@ $\xi=\infty$, $u=0$

$\Rightarrow$

@ $x = -\infty$, $c^- = 0$

@ $x = +\infty$, $c^+ = 0$

@ $\xi_L = +\infty$, $u^+ = 0$

(iii) Laplace transform of dimensionless residence time

$$\bar{f}(s) = \int_0^\infty f(t)\, e^{-st}\, dt$$

$$\mathcal{L}\left(\frac{\partial u}{\partial \tau}\right) = s\bar{u}(s) - u(0)$$

where $u^-(0) = \delta(\xi_L)$

$u^+(0) = 0$

$u(0) = \delta(\xi_L)$

$$\mathcal{L}\left(\frac{\partial u}{\partial \tau}\right) = \frac{\partial \bar{u}(\xi_L, s)}{\partial \xi_L}.$$

⇒ Turning the PDE into ODE's:

$$\frac{d^2\bar{u}^-}{d\xi_L^2} - Pe\,\frac{d\bar{u}^-}{d\xi_L} - Pe\, s\,\bar{u}^- = 0$$

$$\frac{d^2\bar{u}}{d\xi_L^2} - Pe\,\frac{d\bar{u}}{d\xi_L} - Pe\, s\bar{u} = Pe\, \delta(\xi_L) \qquad (\text{for } + \text{ & normal } u)$$

↳ This can be solved numerically

A_Bhadouria_HW11

Tuesday, 11 April 2023 8:47 AM

## Question 1

Rate of absorption of gases A and B based on surface renewal concept

Gases A, B

$y=0$

$y$

$y=\infty$

Liq $A \rightleftharpoons \alpha M \quad (K_A)$

$B \rightleftharpoons \beta M \quad (K_B)$

$$K_A = \frac{k_1}{k_{-1}} = \frac{C_M^{\alpha}}{C_A}$$

$$K_B = \frac{C_M^{\beta}}{C_B}$$

(i) A, B, M have the same diffusion coefficients

Mass balance: $\frac{\partial c}{\partial t} = -\nabla \cdot N + \sum \alpha_i r_i$ ; $r_A = -k_1 C_A + k_{-1} C_M^{\alpha}$ ; $r_M = k_1 C_A - k_{-1} C_M^{\alpha} + k_2 C_B - k_{-2} C_M^{\beta}$

$r_B = -k_2 C_B + k_{-2} C_M^{\beta}$

$$\frac{\partial C_A}{\partial t} = D \frac{\partial^2 C_A}{\partial y^2} + \frac{r_A}{\alpha} = D \frac{\partial^2 C_A}{\partial y^2} + \frac{(-k_1 C_A + k_{-1} C_M^{\alpha})}{\alpha}$$

Similarly,

$$\frac{\partial C_B}{\partial t} = D \frac{\partial^2 C_B}{\partial y^2} + \frac{(k_2 C_B + k_{-2} C_M^{\beta})}{\beta}$$

$$\frac{\partial C_M}{\partial t} = D \frac{\partial^2 C_M}{\partial y^2} + (k_1 C_A - k_{-1} C_M^{\alpha} + k_2 C_B - k_{-2} C_M^{\beta})$$

$$\therefore \quad \alpha \frac{\partial C_A}{\partial t} = \alpha D \frac{\partial^2 C_A}{\partial y^2} - k_1 C_A + k_{-1} C_M^{\alpha}$$

$$\beta \frac{\partial C_B}{\partial t} = \beta D \frac{\partial^2 C_B}{\partial y^2} - k_2 C_B + k_{-2} C_M^{\beta}$$

$$+ \quad \frac{\partial C_M}{\partial t} = D \frac{\partial^2 C_M}{\partial y^2} + k_1 C_A - k_{-1} C_M^{\alpha} + k_2 C_B - k_{-2} C_M^{\beta}$$

$$\frac{\partial}{\partial t}(\alpha C_A + \beta C_B + C_M) = D \frac{\partial^2}{\partial y^2}(\alpha C_A + \beta C_B + C_M)$$

$$\boxed{\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial y^2}}$$

Boundary conditions:

@ $y=0$, $c(0,t) = \alpha C_{A_i} + \beta C_{B_i} + C_{M_i} = C_i$ (Assuming that interfacial conc. is constant throughout time)

@ $t=0$, $c(y,0) = \alpha C_{A_0} + \beta C_{B_0} + C_{M_0}$

@ $y \to \infty$, $c(y \to \infty, t) = \alpha C_{A_0} + \beta C_{B_0} + C_{M_0} = C_0$

(ii) First we solve for the concentration profile:

General solution of PDE's : $c(y,t) = A + B \operatorname{erf}\left(\frac{y}{2\sqrt{Dt}}\right)$

↳ Solving for constants,

$$c(0,t) = c_i = A + B\,\cancel{erf(0)}^{\,0} \Rightarrow A = c_i$$

$$c(\infty,t) = c_0 = A + B\,\cancel{erf(\infty)}^{\,1} = A + B \;;\; B = c_0 - c_i$$

$$\therefore \boxed{c(y,t) = c_i + (c_0 - c_i)\, erf\left(\frac{y}{2\sqrt{Dt}}\right)} \Rightarrow \quad c_A = \frac{c_M^a}{K} \;;\; c_M = c - \alpha c_A - c$$

$$N_A = D\left.\frac{\partial c_A}{\partial y}\right|_{y=0} \;;\; c_A = \frac{c - \beta c_B - c_M}{\alpha} \;;\; N_A = \left.\frac{D\partial c}{\partial y}\right|_{y=0} - \frac{D\beta}{\alpha}\left.\frac{\partial c_B}{\partial y}\right|_{y=0} - \frac{D}{\alpha}\left.\frac{\partial c_M}{\partial y}\right|_{y=0}$$

$$N_B = D\left.\frac{\partial c_B}{\partial y}\right|_{y=0} \;;\; c_B = \frac{c - \alpha c_A - c_M}{\beta} \qquad N_A = \left.\frac{D\partial c}{\partial y}\right|_{y=0} - \frac{\beta}{\alpha}N_B - \frac{1}{\alpha}N_M$$

$$N_M = D\left.\frac{\partial c_M}{\partial y}\right|_{y=0} \;;\; c_M = c - \alpha c_A - \beta c_B$$

$$\therefore \boxed{N_A + \frac{\beta}{\alpha}N_B + \frac{1}{\alpha}N_M = \underbrace{D\left.\frac{\partial c}{\partial y}\right|_{y=0}} = N}$$

$$N = D\left.\frac{\partial c}{\partial y}\right|_{y=0} \;;\; \frac{\partial c}{\partial y} = (c_0 - c_i)\frac{2e^{-y^2}}{\sqrt{\pi t}}$$

$$\left.\frac{\partial c}{\partial y}\right|_{y=0} = (c_0 - c_i)\frac{2}{\sqrt{\pi t}}$$

$$\bar{N} = S \cdot H \int_0^\infty \frac{e^{-st}}{\sqrt{t}}\, dt$$

$$= S \cdot (c_0 - c_i)\frac{2}{\sqrt{\pi}} \cdot D \int_0^\infty \frac{e^{-st}}{\sqrt{t}}\, dt$$

$$\boxed{\bar{N} = \sqrt{DS}\,(c_0 - c_i)}$$

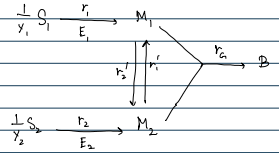
HW12

Tuesday, 18 April 2023 9:38 AM

Bioreactors, cybernetic modelling ; Anagolenetic

Question 1

(i) Product = Biomass ; $M_1, M_2$ → growth precursors ; Glucose, pyruvate → Substrate




Two elementary modes in the network:

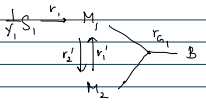
① Diauxic growth : $S_1 \rightarrow M_1$ ; $M_1 \rightarrow M_2$ ; $M_1 + M_2 \rightarrow B$ (Sequential growth)

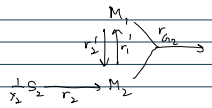
Preferential utilization of $S_1$ over $S_2$ ;

② Simultaneous utilization: $S_1 \rightarrow M_1$ ; $M_1 + M_2 \rightarrow B$

$S_2 \rightarrow M_2$

Two elementary modes are the individual diauxic reactions:




$\frac{1}{Y_1}S_1 \xrightarrow{r_1} M_1$

$M_1 \underset{r_2'}{\overset{r_1'}{\rightleftharpoons}} M_2$

$M_1 + M_2 \xrightarrow{r_{G_1}} B$

Elementary Mode 1

$\frac{1}{Y_2}S_2 \xrightarrow{r_2} M_2$

$M_1 \underset{r_2'}{\overset{r_1'}{\rightleftharpoons}} M_2$

$M_1 + M_2 \xrightarrow{r_{G_2}} B$

Elementary Mode 2

① Mode 1 (Diauxic growth) : $\frac{dc}{dt} = r_{G_1} v_1 + r_{G_2} v_2$

$\frac{ds_1}{dt} = -\frac{1}{Y_1} r_{G_1} Y_1$

$\frac{de_1}{dt} = \alpha_1^* + \frac{k_E S}{K_S + S} u_1 - \beta_e - e_1 (r_{G_1} + r_{G_2})$

$r_{G_1} = \frac{M_1 S_1 c}{K_{S_1} + S_1} \left( \frac{e_1}{e_{1,max}} \right)$

$r_{G_2} = \frac{M_2 S_2 c}{K_{S_2} + S_2} \left( \frac{e_2}{e_{2,max}} \right)$

② Mode 2 ( Simultaneous growth) ; $u_{S_1,M_1} = \frac{Y_{S_1,M_1} r_{S_1,M_1}}{Y_{S_1,M_1} r_{S_1,M_1} + r_{M_2,M_1}}$

(ii) Cybernetic Optimality Laws: ② PSSH → the intermediate concentration is going to zero

Elementary Mode 1

Mass balance on $m_1$ and $m_2$ :

$\frac{dm_1}{dt} = r_1 - r_{G_1} - r_2' + r_1' \approx 0$

$\frac{dm_2}{dt} = r_2' - r_1' - r_{G_1} \approx 0 \Rightarrow r_{G_1} = r_2' - r_1'$

$r_1 - (r_2' - r_1') - r_2' + r_1' = 0$

$\Rightarrow r_1 - r_2' + r_1' - r_2' + r_1' = 0$

$r_1 - 2r_2' + 2r_1' = 0$

$\frac{r_1}{2} = (r_2' - r_1')$

$\therefore \boxed{r_{G_1} = \frac{r_1}{2}}$

Elementary Mode 2

$\frac{dm_1}{dt} = r_1' - r_2' - r_{G_2} \approx 0 \Rightarrow r_{G_2} = r_1' - r_2' \Rightarrow -r_{G_2} = -r_1' + r_2'$

$\frac{dm_2}{dt} = r_2 + r_2' - r_1' - r_{G_2} \approx 0 \Rightarrow r_{G_2} = r_2 + r_2' - r_1'$

$\Rightarrow r_{G_2} = r_2 - r_{G_2}$

$\boxed{r_{G_2} = \frac{1}{2} r_2}$

(iii) Mass balance $E_1$ and $E_2$ : $r_{e_1} = \frac{k_e' S_1}{K_e' + S_1}$ ; $r_{e_2} = \frac{k_e' S_2}{K_e' + S_2}$

$\frac{de_1}{dt} = \alpha_1^* + \frac{k_E' S}{K_e' + S_1} u_1 - \beta_1 e_1 - r_{G_1} e_1$ ($\alpha_1^*$, $r_{G_1}$ → 0)

$\frac{de_2}{dt} = \alpha_2^* + \frac{k_E' S_2}{K_e' + S_2} u_2 - \beta_2 e_2 - r_{G_2} e_2$ ($\alpha_2^*$, $r_{G_2}$ → 0)

$\therefore r_{G_1} = r_{G_1} + r_{G_2}$

Optimality laws:

$u_1 = \frac{r_{G_1}}{r_{G_1} + r_{G_2}}$

$u_2 = \frac{r_{G_2}}{r_{G_1} + r_{G_2}}$

$\boxed{\frac{de_1}{dt} = r_{e_1} u_1 - \beta_1 e_1}$

$\boxed{\frac{de_2}{dt} = r_{G_2} u_2 - \beta_2 e_2}$

$\beta_1 = \beta_2 = \beta_e$

$\frac{dB}{dt} = r_{B_i} V_i$

(iv) $\frac{ds_1}{dt} = -\frac{1}{Y_1} r_1 v_1$ $v_1 = \frac{r_{G_1}}{Max(r_{G_1}, r_{G_2})}$

$$\frac{dB}{dt} = r_{B_i} v_i$$

(iv) $\frac{ds_1}{dt} = -\frac{1}{Y_1} r_1 v_1$ $\qquad v_1 = \frac{r_{a_1}}{\text{Max}(r_{a_1}, r_{a_2})}$

$\frac{ds_2}{dt} = -\frac{1}{Y_2} r_2 v_2$ $\qquad v_2 = \frac{r_{a_2}}{\text{Max}(r_{a_1}, r_{a_2})}$

We also know, $r_1 = 2r_{a_1}$

$r_2 = 2r_{a_2}$

$$\frac{ds_1}{dt} = \frac{2r_{a_1} v_1}{Y_1} = -\frac{1(2)}{Y_1} \frac{r_{a_1}^2}{\text{Max}(r_{a_1}, r_{a_2})}$$

$$\frac{ds_2}{dt} = \frac{2r_{a_2} v_2}{Y_2} = -\frac{1(2)}{Y_2} \frac{r_{a_2}^2}{\text{max}(r_{a_1}, r_{a_2})}$$

Mass balance on B:

$$\frac{dB}{dt} = r_{a_1} v_1 + r_{a_2} v_2$$

$$r_1 = \frac{k_1 s_1 e_1}{K_1 + s_1} = 2r_{a_1}$$

$$r_2 = \frac{k_2 s_2 e_2}{K_2 + s_2} = 2r_{a_2}$$

(v) Cybernetic variables $u_i$ and $v_i$ expressed earlier

$u_1 = \frac{r_{a_1}}{r_{a_1} + r_{a_2}}$ $\qquad v_1 = \frac{r_{a_1}}{\text{Max}(r_{a_1}, r_{a_2})}$

$u_2 = \frac{r_{a_2}}{r_{a_1} + r_{a_2}}$ $\qquad v_2 = \frac{r_{a_2}}{\text{Max}(r_{a_1}, r_{a_2})}$

Respective relationships for $r_g$ and $r_{a_2}$ can be plugged in

(vi) Code attached.

# HW13

Tuesday, 25 April 2023 3:22 PM

$2A \xrightarrow{k} \text{Products}$ ; Dispersed phase mixing
Occurring only on the drops of dispersed ; Second order
Liq–Liq dispersion

Cont., CSTR ; $\theta = \frac{\omega N_0 V}{q}$

(i) $\mu_n(\tau) = \int_0^1 x^n \phi(x,\tau)\cdot dx$ — dimensionless population density

$\mu_0(\tau) = \int_0^1 x^0 \phi(x,\tau)\cdot dx = \int_0^1 \phi(x,\tau)\cdot dx = 1$

(ii) Steady state reactor, CSTR population balance:

$\frac{\partial n(c,t)}{\partial t} + \frac{\partial [\dot{C} n]}{\partial c} = h^+ - h^- + [\text{IN}] - [\text{Out}]$ ; $h^- = \omega n$

SS

Coalescence – redispersion

$h^+ = 2\omega \iint n(c_1,t)\, n(c_2,t)\, dc_1\, dc_2$

$\dot{C} = -r(c) = -kc^2$

$\frac{\partial [\dot{C} n]}{\partial c} = \omega \iint_{c_1+c_2=c} n(c_1,t)\, n(c_2,t)\, dc_1\, dc_2 + \frac{q}{V}\left(\delta(c-c_0)\right) - \left(\frac{q}{V} n\right)$

$\times \frac{x^r \cdot c_0}{\omega N_0}$

$\frac{x^r c_0}{N_0}\frac{\partial [-kc^2 \cdot n]}{\partial c} = \frac{x^r c_0}{N_0} \iint_{c_1+c_2=c} n(c_1,t)\, n(c_2,t)\, dc_1\, dc_2 + \frac{q}{V}\delta(x-x_0)\cdot\frac{x^r c_0}{N_0} - \frac{x^r c_0}{\omega N_0}\left(\frac{q}{V} n\right)$

Taking integral $\int_0^1$ & non-dimensionalizing based on class notes:

$n \to \phi$

$x^n \phi \to \mu_n$

[In] term: $\int_0^1 \frac{q}{V} \frac{c_0}{N_0} \frac{\delta(c-c_0)}{\omega}\cdot dx = \frac{q}{\omega V N_0}\int_0^1 \delta(x-x_0)\cdot dx = \frac{1}{\theta}$

[Out] term: $\int_0^1 \frac{x^r c_0}{N_0 \cdot \omega}\cdot\frac{q}{V} n \cdot dx = \frac{q}{\omega V N_0}\int_0^1 x^r \cdot \left(\frac{c_0 n}{N_0}\right)\cdot dx = \frac{1}{\theta}(\mu_n)$ — $\phi(x,\tau)$

$h^+$ term: $x^r c_0 \left[1 \quad \cdots \quad 1 \right] \leq \frac{n!}{\quad} \left[\int_0^1 x^{n-r} \quad \int_0^1 x^r \right]$

$r^{th}$ term : $\frac{x^r c_0}{N_0}\int n_{c_1} n_{c_2}\, dc_1\, dc_2 = \frac{1}{2^n}\sum_{r=0}^{n}\frac{n!}{(n-r)!\,r!}\int_0^1 x^{n-r}\,dx \int_0^1 x^r \phi \cdot dx$

(Using Binomial theorem)

$n^{th}$ term : $\int_0^1 x^n \phi \cdot dx$

$\frac{\partial[-kc^2 n]}{\partial c}$ : $\frac{\partial(-kc^2 n)}{\partial(c_0 x)} = \frac{k}{c_0}\frac{\partial c^2}{\partial x}\cdot n - \frac{k c^2}{c_0}\frac{\partial n}{\partial x} = \frac{k c_0 n}{\omega N_0}\int_0^1 x^{n+1}\phi \cdot dx$

$$\frac{k c_0 n}{\omega N_0}\int_0^1 x^{n+1}\phi\cdot dx - 2^{-n}\sum_{r=0}^{n}\frac{n!}{(n-r)!\,r!}\int_0^1 x^{n-r}\phi\cdot dx\int_0^1 x^r\phi\cdot dx + \int_0^1 x^n\phi\, dx = \underbrace{\frac{q}{\omega N_0 V}\left(1-\int_0^1 x^n\phi\cdot dx\right)}_{\text{In - out}}$$

$$\frac{k c_0 n}{\omega N_0}\int_0^1 x^{n+1}\phi\cdot dx = \frac{q}{\omega N_0 V}\left(1-\int_0^1 x^n\phi\cdot dx\right) - \int_0^1 x^n\phi\, dx + 2^{-n}\sum_{r=0}^{n}\frac{n!}{(n-r)!\,r!}\int_0^1 x^{n-r}\phi\cdot dx\int_0^1 x^r\phi\cdot dx$$

$$K n \mu_{n+1} = \frac{1}{\theta}(1-\mu_n) - \mu_n + 2^{-n}\sum_{r=0}^{n}\frac{n!}{(n-r)!\,r!}\mu_{n-r}\mu_r$$

(iii) $n=1$ ;

$$K\mu_2 = \frac{1}{\theta}(1-\mu_1) - \mu_1 + 2^{-1}\left(\frac{1!}{1!\,0!}\mu_1\mu_0 + \frac{1!}{1!\,0!}\mu_0\mu_1\right)$$

$$K\mu_2 = \frac{1}{\theta}(1-\mu_1) - \cancel{\mu_1} + \frac{1}{\cancel{2}}\left(\cancel{2}\mu_1\right)$$

$$\boxed{\mu_2 = \frac{1}{K\theta}(1-\mu_1)}$$

(iv) $\mu_{n+1} \to 0$ for large $n$ ; $\theta \gg 1$

$$\mu_n = \int_0^1 x^n \phi(x,\tau)\cdot dx$$

$n \to \infty$ $\quad x^n \to 0$ since $x$ is a non-dimensional value b/w 0 & 1

$\therefore \mu_n \to 0 \quad \text{as } n \to \infty$

(v) $K = \{0.1, 1, 10\}$

Have to solve for $\mu_1$ ; $\mu_2 = \frac{1}{K\Theta}(1-\mu_1)$

↳ Have to iteratively solve until large $n$ values where $\mu_n$ for large $n$ is equal to 0.

$\mu_0 = 4$
$\mu_1 = f(\mu_2)$
$\mu_2 = f(\mu_3)$
⋮
$\mu_{10} = 0$

# Lecture notes

Monday, 16 January 2023 4:16 PM

# Lecture 1

Saturday, 14 January 2023 9:24 PM

Lecture 1

$$R = \frac{dX}{dt}$$

$$r = \frac{d\xi}{dt}$$

$$\xi = \frac{X}{V}$$

$$r \times V = \frac{dX}{dt}$$

$$\frac{N_j - N_{j0}}{\alpha_j} = X$$

$$\downarrow$$

$$\frac{dN_j}{dt} = \alpha_j \frac{dX}{dt}$$

$$= \alpha_j V r$$

$$N_j = c_j V$$

$$\frac{d(c_j V)}{dt} = \alpha_j r V$$

$$\downarrow$$

chain rule

$$\downarrow$$

$$\frac{dc_j}{dt} = \alpha_j r - c_j \frac{d \ln V}{dt}$$

$$\left\{ \frac{1}{V}\frac{dV}{dt} = \frac{d}{dt} \ln V \right\}$$

$$\frac{dv_j}{dt} = \alpha_j r - c_j \frac{d \ln V}{dt}$$

in terms of $\varphi_k$ $\left( \varphi_k = \frac{x}{V} \right)$

$$\frac{dx}{dt} = \frac{d}{dt}(\varphi_k V) = rV$$

$$\therefore \quad \frac{d\varphi_k}{dt} = -\varphi_k \frac{d \ln V}{dt} + r$$

# Lecture 2

Tuesday, 17 January 2023 12:00 PM

$c_j = \frac{\rho_j}{m_j}$ ; $J_j = \frac{j_j}{m_j}$ = molar flux relative to $v$

perfect mixing → Don't need momentum balance

Imperfect mixing → Residence time distribution

Continuity eq:

$$\frac{\partial c_j}{\partial t} + v \cdot \nabla c_j + \nabla \cdot J_j + c_j(\nabla \cdot v) = \sum \alpha_{ij} r_i$$

→ Rowing across the river (fishing) → gives total derivative (material derivative)

→ Fishing at the spot → Local derivative

## Heterogenous reactions

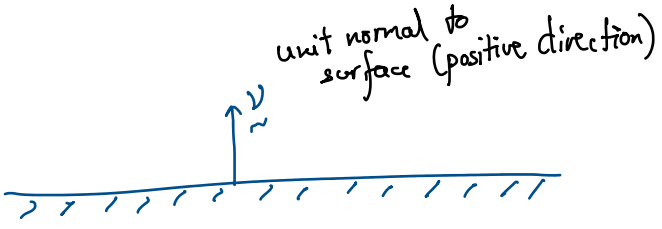
$$\frac{Dc_j}{Dt} + \nabla \cdot J_j = \sum_j^m \alpha_{ij} a_v \hat{r}_i - c_j \frac{D \ln \hat{V}}{Dt}$$

Transient term; Diffusive term

$a_v$ = active per unit volume

$\hat{r}$ = reaction per unit area of catalyst surface.

Reaction located at a specific part → occurs at a boundary




Intrinsic reaction rate is always positive

$\alpha_j > 0$ (for reactant)

↳ going to a catalyst surface

rate at which the reaction is happening

positive

↳ going to a catalyst surface

rate at reaction is happening

Consuming solid surface

$\partial\Omega_t \rightarrow \partial\Omega_{t+dt}$

↑ shape of the surface at time $t$

$x - \frac{\nabla f(x,t)}{\|\nabla f(x,t)\|} \hat{r}\, dt$

Leading to change in surface shape

Normal to a surface

Represents a location on the solid surface

## Chemical kinetics:

→ Mass action kinetics ⇒ (of elementary reactions)

$A \rightarrow B$

$r = k[C_A]^1$

→ Homogenous reaction: + deDonder kinetics

$\sum_{j=1}^{s} \alpha_j A_j = 0$ ← given reaction sequences.

Batch: $c_j = c_{j_0} + \alpha_j \xi$ (No change in volume)

How the reaction rate changes as $\xi$ changes

$$r(c_j, T) = k \prod_{j=1}^{s} (c_{j_0} + \alpha_j \xi)^{a_j} - k' \prod_{j=1}^{s} (c_{j_0} + \alpha_j \xi)^{a'_j}$$

reverse reaction

$$\underset{\sim}{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_j \end{bmatrix}$$

$$\left.\frac{\partial r}{\partial \xi}\right|_T = ?$$

We define:

$$a_j \equiv \frac{1}{2}(|\alpha_j| - \alpha_j) \quad ; \quad \alpha_j < 0 \text{ (reactant)} \Rightarrow a_j = -\alpha_j$$

$|\alpha_j| = -\alpha_j$

or

$a_j = 0$ (if $\alpha_j \geq 0$)

$a_j = 0$ $\alpha_j$ [illegible]

Similarly:

$$a_j' = \frac{1}{2}\left(|\alpha_j| + \alpha_j\right) = \begin{cases} 0 & \alpha_j < 0 \\ \alpha_j & \alpha_j > 0 \end{cases}$$

Now we differentiate:

$$\left.\frac{\partial r}{\partial \xi}\right|_T = \sum_{r=1}^{S} \frac{\partial c_r}{\partial \xi} \frac{\partial r}{\partial c_r}$$

$$= \sum_{r=1}^{S} k a_r \left( \prod_{j=1}^{S} c_j^{a_j} \right)$$

# Lecture 3

Thursday, 19 January 2023 12:03 PM

Lecture 3

Lecture 4

Tuesday, 24 January 2023 12:01 PM

## Intrinsic Kinetics in Catalysis

Active centers ; Homogenous catalysis → Active centers in the bulk

Heterogenous catalysis → " on surface

$A + B \overset{Cat.}{\rightleftharpoons} C + D$

$A + X_1 \rightleftharpoons AX_1$
$B + X_2 \rightleftharpoons BX_2$
} Chemisorption

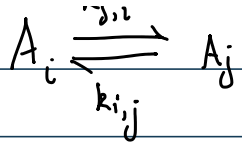
$AX_1 + BX_2 \rightleftharpoons C + D + X_1 + X_2$ } Surface reaction & desorption

$P + f X_{1,2} \rightleftharpoons 0$
$f' X_{1,2} \rightarrow \bar{X}_{1,2}$
} Poisoning (Thermal deactivation)

## Lumping in Kinetic Systems

Petroleum → lumped into 4 major components: Paraffins, Olefins, Naphthalene, Aromatics } PONA (Also called "Naphtha")

$A_i \underset{k_{i,j}}{\overset{k_{j,i}}{\rightleftharpoons}} A_j$




$\sum_{i=1,\, i \neq j}$ ; $A_i$ going to all different species (consumed will have →)

Reducing no. of species to deal w/ ⟶ Lumping

# Lecture 7

Tuesday, 31 January 2023 12:02 PM

$$\frac{d\underset{\sim}{x}}{dt} = \underset{\sim}{K}\underset{\sim}{x} \quad ; \quad \underset{\sim}{K} \equiv \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1n} \\ \vdots & & \ddots & \\ k_{n1} & & & k_{nn} \end{bmatrix} \quad ; \quad k_{ii} = -\sum_{j}^{n} k_{ji}$$

$\underset{\sim}{K}\underset{\sim}{x} = \underset{\sim}{0}$ ; $k_{ij} x_j^* = k_{ji} x_i^*$ → principle of microscopic reversibility

$$\sqrt{\underset{\sim}{D}} = \begin{bmatrix} \sqrt{x_1^*} & \cdots & 0 \\ & \sqrt{x_2^*} & \\ \vdots & & \ddots \\ 0 & & \sqrt{x_n^*} \end{bmatrix} \quad ; \quad \left(\sqrt{\underset{\sim}{D}}\right)^{-1} \underset{\sim}{K} \left(\sqrt{\underset{\sim}{D}}\right) = \underset{\sim}{K}$$

$\underset{\sim}{B}^{-1}\underset{\sim}{A}\underset{\sim}{B} = \underset{\sim}{C}$ ; A & C have the same

$\underset{\sim}{B}^{-1}\underset{\sim}{A}\underset{\sim}{B} - \lambda\underset{\sim}{B}^{-1}\underset{\sim}{B} = \underset{\sim}{C} - \lambda I$

$\underset{\sim}{B}^{-1}(\underset{\sim}{A} - \lambda\underset{\sim}{I})\underset{\sim}{B} = \underset{\sim}{C} - \lambda\underset{\sim}{I}$

$|(A - \lambda I)| = |C - \lambda I| = 0$

$\underset{\sim}{z}^{\dagger}\underset{\sim}{K}\underset{\sim}{z} \leq 0 \qquad z_j \underset{\sim}{D}^{-1} z_k = 0 \qquad j \neq k$

$$\underline{x} = \sum_{j=0}^{n-1} \alpha_j \underline{z}_j \quad ; \quad \underline{z}_0 = \underset{\sim}{x}^*$$

$$\left(\sqrt{D}\right)^{-1} K \left(\sqrt{D}\, z_j\right) = \tilde{\underset{\sim}{K}}\, \tilde{z}_j \quad ; \text{ given } z_j \underset{\sim}{D}^{-1} z_k = 0$$

We can find $\alpha_j$ : $\boxed{\alpha_j = \dfrac{\underset{\sim}{z}_j \underset{\sim}{D}^{-1} \underset{\sim}{x}}{\underset{\sim}{z}_j \underset{\sim}{D}^{-1} \underset{\sim}{z}_j}}$

$$\underset{\sim}{x}_0 = \sum_{k=0}^{n-1} \frac{(\underset{\sim}{x}_0 \underset{\sim}{D}^{-1} \underset{\sim}{z}_k)}{(z_k \underset{\sim}{D}^{-1} \underset{\sim}{z}_k)} \underset{\sim}{z}_k e^{\lambda_k t}$$

## Parameter Estimation in Nonlinear Kinetics

Determining kinetic parameters : $r \rightarrow$ rate

$\xi \rightarrow$ extent

$k \rightarrow$ rate constants $k = [k_1\ k_2 \cdots k_p] \rightarrow p$ parameters

$\frac{d\xi}{dt} = r(\xi, k)$ ; $\xi(0) = 0$ → ODE representing the temporal evolution of $\xi$

$\{\hat{\xi}_i\}_{i=1}^{N}$ : $\underset{k \in \mathbb{R}^p}{Min} = \sum_{i=1}^{N} |\xi_i(t,k) - \hat{\xi}_i|^2$

prediction vs. measured

Bellman's elegant quasi-linearization approach

$m^{th} \rightarrow (m+1)^{th}$

$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ ; $n = p+1$ ; $x_1 = \xi$ , $x_2 = k_1, \ldots, x_n = k_p$ (Redefining the variables)

$f_1(x) = r(\xi, k)$ ; $f_2(x) = 0$ . ⇒ $\underline{f}$

→ $\frac{d\underline{x}}{dt} = \underline{f}(x)$ → New ODE

Jacobian (derivative w.r.t each variable)

Linearize this → $\frac{dx^{(m+1)}}{dt} = \underline{f}(x^{(m)}) + (\nabla \underline{f}(x^{(m)}))(x^{(m+1)} - x^{(m)})$ ; Let $J^{(m)} \equiv \nabla \underline{f}(x^m)$

↳ particular soln: $\frac{dp}{dt} = \underline{f}(x^{(m)}) + J^{(m)}(p - x^{(m)})$ ; $p(0) = 0$

↳ Homogeneous soln: $\frac{dh^j}{dt} = J^{(m)} h^{(j)}$ ; $h^j(0) = e^j \equiv [0 \ldots 1 \ 0 \ldots 0]^T$

$\therefore x^{(m+1)}(t) = p(t) + \sum_{j=1}^{n} x_j(0) h^j(t)$ ; p unknowns $\{x_j(0)\}$

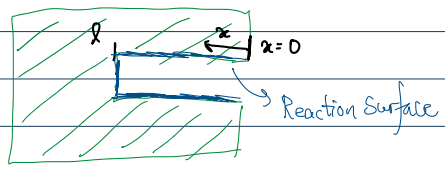
Minimize first component → extent of reaction

$\sum_{i=1}^{N} \left[ p_1(t_i) - \sum_{j=1}^{n} x_j(0) h^j(t_i) - \hat{\xi}_i \right]^2$ → Iterate until satisfactory convergence.

## Transport Effects in Catalytic Reaction Systems

Single Pore Analysis:

+ Steady state
+ First order
+ Isothermal



$$-DA\left.\frac{dc}{dx}\right|_x - \left(-DA\left.\frac{dc}{dx}\right|_{x+dx}\right) + \hat{K}_c P dx = 0$$

Diffusive flux

$\hat{K}_c P dx$ ↓ reaction

BC's: 1) @ $x=0$, $c=c_0$

2) @ $x=l$, $\frac{dc}{dx}=0$

$$\therefore D\frac{d^2c}{dx^2} + \frac{P}{A}\hat{K}_c = 0$$

$0 < x < l$

Whatever diffuses in the pore reacts: so $DA\left.\frac{dc}{dx}\right|_{x=0}$ → Total reaction rate in pore

Reaction rate in pore at uniform conc: $c_0 = Pl\hat{K}c_0$

Effectiveness factor Matrix, $N_i$: $NPl\hat{K}c_0 = \left[DA\frac{dc}{dx}\right]_{x=0}$

Non-dim: $\xi = \frac{x}{l}$ ; $\phi^2 = \frac{-Pl^2D^{-1}\hat{K}}{A}$

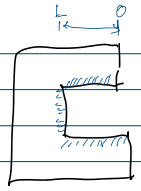
$$\therefore \frac{d^2c}{d\xi^2} = \phi^2 c$$

$c = [\cosh\phi(1-\xi)]k_1 + [\sinh\phi(1-\xi)]k_2$ ; $k_1$ & $k_2$ are constants

Lecture 8

Thursday, 2 February 2023 12:02 PM

$$\underset{\sim}{D}\frac{d^2\underset{\sim}{c}}{dx^2} + \frac{P}{A}\hat{\underset{\sim}{K}}\underset{\sim}{c} = 0 \quad ; \quad 0 < x < l$$



$$l = \underset{\sim}{D}A \left.\frac{dc}{dx}\right|_{x=0}$$

$$c_0 = \frac{P}{\hat{K}c_0}$$

$$NPl\hat{K}c_0 = \left[\underset{\sim}{D}A\frac{dc}{dx}\right]_{x=0}$$

Non-dimensionalizing : (Positive definite w/ the negative sign) Making

$$\xi = \frac{x}{l} \quad ; \quad \phi^2 = \frac{-Pl^2\underset{\sim}{D}^{-1}\hat{K}}{A}$$

$$\frac{d^2c}{d\xi^2} = \phi^2 c \Rightarrow c = a'\cos\phi\xi + b'\sinh\phi\xi = a\cosh\phi(1-\xi) + b'\sinh\phi(1-\xi) \quad ; \text{ BC: } \frac{dc}{d\xi} = 0 \text{ @ } \xi = 1$$

$$c_0 = (\cosh\phi)h_1 \text{ @ } \xi = 0$$

Sylvester's decomposition/expansion of a matrix:

$$\underset{(n\times n)}{\underset{\sim}{A}} = \sum_{j=1}^{n}\lambda_j\underset{\sim}{P}_j$$

$$\underset{\sim}{P}_j = \frac{\text{adj}(\lambda_j\underset{\sim}{I} - \underset{\sim}{A})}{\prod_{r=1,j}^{n}(\lambda_j - \lambda_r)} \rightarrow \text{projection matrix} \qquad \underset{\sim}{P}_j^2 = \underset{\sim}{P}_j \quad (\underset{\sim}{P}_i\underset{\sim}{P}_j = 0 \quad i \neq j)$$

$(\lambda_1, \lambda_2 \ldots \lambda_n)$

Let $\underset{\sim}{w}_1^T, \underset{\sim}{w}_2^T, \ldots \underset{\sim}{w}_n^T$ be the row eigenvectors of $\underset{\sim}{A}$

Let $\underset{\sim}{v}_1, \underset{\sim}{v}_2, \ldots \underset{\sim}{v}_n$ be the column eigenvectors of $\underset{\sim}{A}$

$$\underset{[1\times n]}{\underset{\sim}{w}_j^T}\underset{(n\times n)}{\underset{\sim}{A}} = \lambda_j\underset{\sim}{w}_j^T \quad ; \quad \underset{\sim}{A}\underset{\sim}{v}_j = \lambda_j\underset{\sim}{v}_j$$

$$\underset{\sim}{P}_j = \frac{\underset{\sim}{v}_j\underset{\sim}{w}_j^T}{\underset{\sim}{w}_j^T\underset{\sim}{v}_j} \qquad \underset{\sim}{w}_j^T\underset{\sim}{v}_k = 0 \quad j \neq k$$

Biorthogonality

$$\underset{\sim}{P}_j^2 = \frac{(\underset{\sim}{v}_j\underset{\sim}{w}_j^T)(\underset{\sim}{v}_j\underset{\sim}{w}_j^T)}{(\underset{\sim}{w}_j^T\underset{\sim}{v}_j)(\underset{\sim}{w}_j^T\underset{\sim}{v}_j)} = \frac{\underset{\sim}{v}_j(\underset{\sim}{w}_j^T\underset{\sim}{v}_j)\underset{\sim}{w}_j^T}{(\underset{\sim}{w}_j^T\underset{\sim}{v}_j)(\underset{\sim}{w}_j^T\underset{\sim}{v}_j)} = \underset{\sim}{P}_j$$

$$\underset{\sim}{c}^A = \sum_{j=1}^{N} e^{\lambda_j}\underset{\sim}{P}_j$$

$$\cosh\underset{\sim}{\phi} = \sum_{j=1}^{n}\cosh\lambda_j\underset{\sim}{P}_j \quad ; \quad \underset{\sim}{\phi}^{-1} = \sum\lambda_j^{-1}\underset{\sim}{P}_j$$

$$[\cosh\underset{\sim}{\phi}\underset{\sim}{J}]^{-1} = \underset{\lambda=0}{\underset{\sim}{P}_j} + \sum_{j=2}^{n}(\cosh\lambda_j)^{-1}\underset{\sim}{P}_j$$

(since $\cosh 0 = 1$)

Temperature Effect on Apparent Activation Energy

# Temperature Effect on Apparent Activation Energy

$\ln k$ vs. $\frac{1}{T}$ ; if no transport effects : straight line $\frac{E}{R}$

However, diffusion limitations distort this behaviour

$\hat{r}_{obs} = \hat{r}\eta$ (assuming uniform temp.)

$\hat{r} = Ae^{-E/RT} \cdot c_s$ ; $c_s$ = reactant

@ fixed $c_s$: $\frac{d\ln \hat{r}}{dT^{-1}} = \frac{d\ln \hat{k}}{dT^{-1}} = \frac{E}{R}$

# Lecture 9

Tuesday, 7 February 2023 12:02 PM

$$\nabla^2 u = \phi^2 u \quad \text{in } \Omega \quad ; \quad u = 1 \text{ on } \partial\Omega$$

$$u = \sum_{n=0}^{\infty} \phi^{2n} u_n(p)$$

$$\nabla^2 u = \sum_{n=0}^{\infty} \phi^{2n}$$

Lecture

Tuesday, 21 February 2023 12:02 PM

**Steady State** (Non-isothermal) $\tilde{v}(\rho)$ ; $\nabla^2 v = -\beta\phi^2 G_1(\tilde{v})$ $\quad u = \frac{1+\beta-v}{\beta}$ ← reducing dimensionality (don't have to care about $\beta$)

$v = \tilde{v} + y$ (perturbation) ↳ We want to see if

$$\begin{cases} -\nabla^2 y + g(\rho) y = \frac{\partial y}{\partial t} \\ \frac{\partial y(0,\tau)}{\partial \rho} = 0 \quad ; \quad y(1,\tau) = 0 \end{cases} \quad ; \quad g(\rho) = -\beta\phi^2 G_1'(\tilde{v}(\rho)) \leftarrow \text{Linearized problem}$$

Initial boundary value problem

solving eigenvalue problem (using BC & orthogonality)

$$y(\beta,\tau) = \sum_{j=1}^{\infty} a_j z_j(\rho)\, e^{-\mu_j \tau}$$

Perturbation solution

→ $a_j$, $z_j$ are still unknown

$\mu_j$ cannot be negative (or else it will blow up)

If $\mu_j < 0$, $\tilde{v}$ is unstable

**Sturm Comparison Theorem** (Looking for instability) (We want unstability so $\mu$ should be negative)

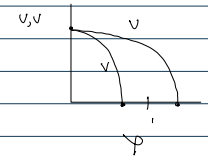
Eigenvalue problem: $-\nabla^2 u + g(\rho)u = +\mu u$

BC: $u'(0) = 0$, $u(1) = 0$ } homogeneous BC

Given an IVP: $V(\rho)$,

$-\nabla^2 V(\rho) + g(\rho)V = 0 \quad 0 < \rho < 1$ } $-\nabla^2 v + \underbrace{g(\rho)}_{\text{This dictates how much it goes up \& down}} v = \mu v$

$V(0) = 1, \quad V'(0) = 0$ } $v(0) = 1, \quad v'(0) = 0$




$V$ & $v$ have the same no. of zeros

According to Sturm → given, $i^{th}$ zero of $V$ → $i^{th}$ zero of $v$ will be on the right.

∴ Suppose $V$ has a zero in $(0,1)$; then there must be a negative $\mu$ s.t. $v(1) = 0$ (i.e. $v$ has a root at 1) } This is different from a BC since it applies to the solution

↳ This marks the limit of stability for $\tilde{v}$

↳ So then now we need condition for $V$ which allows it to have a zero in $(0,1)$

# Past exam 2021

Wednesday, 22 February 2023 1:39 PM

Q1] $2A_1 + 3A_2 \rightarrow 2A_3$ No residual amount of $A_1, A_2$ & $A_3$

$A_1 + A_2 + A_3 \rightarrow 3P$

$A_1 + \frac{3}{2}A_2 \rightarrow A_3$

$A_1 + A_2 + A_3 \rightarrow 3P$

$A_1 + A_2 + (A_1 + \frac{3}{2}A_2) \rightarrow 3P$

$2A_1 + \frac{5}{2}A_2 \rightarrow 3P$ $\quad \frac{A_1}{A_2} = \frac{2}{5/2} = \frac{4}{5}$

$2A_1 + 3A_2 - 2A_3 = 0$

$A_1 + A_2 + A_3 - P = 0$

$$\begin{bmatrix} 2 & 3 & -2 & 0 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

Solution:

$dN_j = \alpha_j dX$ $\quad X \rightarrow$ extent of reaction } for 1 reaction

$\Delta N_1 = N_{1f} - N_{1i} = -2X_1 - X_2$ $\quad X = \frac{N_f - N_i}{\alpha}$

$\Delta N_2 = -3X_1 - X_2$

$\Delta N_3 = 2X_1 - X_2$

$\Delta N_4 = 3X_2$

$$\underline{\Delta N} = \underline{A}\,\underline{X}$$

$$\Delta \begin{bmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ -3 & -1 \\ 2 & -1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$\underline{A}^T = \begin{bmatrix} -2 & -3 & 2 & 0 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$ Wide matrix (max rank = 2) — Has a null space

$\underline{A}^T z = 0$

So $\underline{A}^T z = 0$ → must also have a non-trivial solution

$$z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}$$

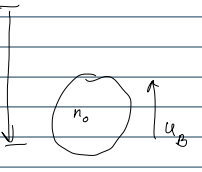
$\underline{A}^T z = 0$ $\quad -2z_1 - 3z_2 + 2z_3 + 0z_4 = 0$ } $-2z_1 - 3z_2 + 2z_3 = 0$

$-1z_1 - z_2 - z_3 + 3z_4 = 0$ } (-1) $2z_1 + 2z_2 - 2z_3 + 6z_4 = 0$

$-z_2 + 4z_3 - 6z_4 = 0$

(+) $-2z_1 - 5z_2 + 6z_4 = 0$

Q2] $2A \rightarrow B$ Isothermal, Ideal gas

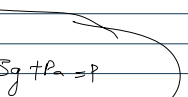



→ Bubble volume at time t.

(a) Production of gas → rate $\quad r = k[A]^2$

$V = \frac{NRT}{P}$

Pressure balance: $P = \rho g h_0$ $\quad P = \rho g (u_B \cdot t)$




$(H_0 - u_B t)\rho g + P_a = P$

Mass balance: $dN_j = \alpha_j dX$

Mass balance : $dN_j = \alpha_j \, dx$

$$\frac{dN_j}{dt} = \alpha_j \frac{dx}{dt} = \alpha_j V \cdot r \qquad r = k[A]^2$$

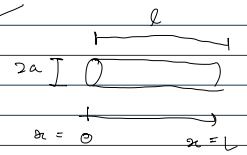
$$\frac{dN_A}{dt} = -2V \cdot r \qquad V = \frac{NRT}{P}$$

$V(t)$

$$\frac{dN_A}{dt} = -2\frac{NRT}{P} r$$

$$\alpha_j \frac{dx}{dt} = \alpha_j V \cdot r$$

Q3]




$\hat{V}_{max}$ = max. area-specific reaction rate

pore-diffusion problem

(i) Shell balance :

$$D\frac{d^2c}{dx^2} + \frac{P}{A}\hat{r}(c) = 0$$

BC: @ $x = 0$, $c = c_s$

@ $x = l$, $\frac{dc}{dx} = 0$

$$\frac{l \cdot 2\pi a}{\pi a^2} = \frac{2l}{a}$$

$$D\frac{d^2(c_s u)}{d(\rho l)^2} = -\hat{r}(c) \cdot \frac{2l}{a}$$

$$\hat{r}(c) = R(u)\hat{r}(c_s)$$

$$\frac{Dc_s}{l^2}\frac{d^2u}{d\rho^2} = -\hat{r}(c) \cdot \frac{2l}{a}$$

$$\frac{d^2u}{d\rho^2} = -R(u)\hat{r}(c_s)\frac{l^2}{Dc_s} \cdot 2 \cdot l$$

$$\frac{d^2u}{d\rho^2} = -\left(\frac{2l^2 \cdot \hat{r}(c_s)}{a\, D_{c_s}}\right) \cdot R(u) \cdot l$$

$$\frac{d^2u}{d\rho} = -\phi^2 \cdot R(u)\, l$$

$$R(u) = \frac{\hat{r}(c)}{\hat{r}(c_s)} = \frac{\frac{\hat{V}_{max} c}{K + c}}{\frac{\hat{V}_{max} c_s}{K + c_s}} = \frac{c}{K+c}\frac{(K+c_s)}{c_s} = \frac{c}{c_s}\frac{(K+c_s)}{(K+c)}$$

$$= \frac{u(K + c_s)}{(K+c)} = \frac{u}{c_s}\frac{(K+1)}{(K+\frac{c}{c_s})}$$

$$= \frac{\frac{u}{c_s}(K'+1)}{\frac{1}{c_s}(K' + \frac{c}{c_s})}$$

$$= \frac{u(K'+1)}{(K'+u)}$$

Lecture:Biomass

Tuesday, 28 March 2023 12:09 PM

→ Cybernetic models: Pyruvate - Fumerate

→ Hybrid cybernetic model → metabolic networks
(PSSH for intracellular metabolites)

www.expasy.org (Structure of metabolic networks)

→ Elementary mode

→ Linear programming → deal w/ max. a linear functional
maximizing
↳ no of feasible soln are on a convex polygon
(solutions are hopping from one vertex to another)

Choosing which path
(by metabolizing)
↓
maximizing the
final output
(Total carbon intake, or max. viable etc.)

Convex comb.
↳ Linear comb. w/ positive coefficients

↓
Flux space and yield space

→ FBA solution vs. Cybernetic model
(Paulsen)

→ $n(m,t)$ ← Total number density

No density ← $N(t) = \int_0^\infty n(m,t)\,dm$ ← All cells in the interval

→ $r(m,s)$

↑
Rate of increase of cell mass

Flux = $r(a,s) \cdot n(a,t)$

→ Division rate of a cell:
($\Gamma(m,s)$) → Fraction of cells of mass m dividing

→ Daughter cell Mass dist: [contribute differently to the mass]

$p(m|m',s)\,dm$ → fraction of daughter cells formed from mother cells of mass $m'$ (w/ mass b/w $m$ & $m+dm$)

↑
given a known certain event
(independent vs. dependent event)

**Population balance Equation:**

no. of cells of mass m b/w a & b

$\frac{d}{dt}\int_a^b n(m,t)\,dm$ = Rate of increase of cells of mass m

m and t are independent vars (can take $\frac{d}{dt}$ inside)

= Rate of cells entering at a − Rate leaving @ b − rate of cells lost from (a,b) from division + Rate of cells arriving by division of larger cells

$r(a,s)\cdot n(a,t)$ | $r(b,s)\,n(b,t)$ | $\int_a^b \Gamma(m,s)\,n(m,t)\,dm$ | $\int_a^b dm \int_m^{2'} \Gamma(m',s)\,n(m',t)\,dm'\,p(m|m',s)$

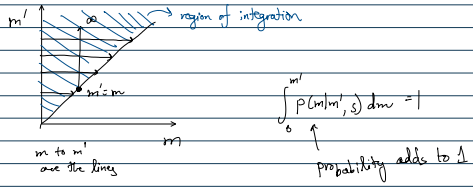
2 cells

r @ m=0 is 0

at any given time $n \neq \infty$ ; i.e @ $m=\infty$, $n=0$

r @ $m=0$ is 0

at any given time $n \neq \infty$ ; i.e. @ $m=\infty$, $n=0$




$$\int_0^{m'} p(m|m',s)\,dm = 1$$

↑ probability adds to 1

$$\therefore \frac{dN}{dt} = \int_0^{\infty} I(m,s)\, n(m,t)\, dm$$

$$c(t) = \int_0^{\infty} m \cdot n(m,t)\, dm$$

↑ Concentration (biomass)

+ Biomass balance for lumped biophase:

$$+ \frac{ds}{dt} = -\frac{1}{r}\int_0^{\infty} dm\; n(m,t)\, r(m,s)$$

Dissolution process

$Y$ → conc. of solute (in bulk)

$Y^*$ → solubility

$$\frac{dN}{dt} = -k' \lim_{m\to\infty}\left[m^{2/3}\, n(m,t)\right]$$

↑ negative because losing mass

Internal coordinate change → due to reaction

$S(x)$ → differential interface surface.

Method of characteristics [hyperbolic vs. elliptical?]

2 cells from each method

Lecture population balances

Tuesday, 4 April 2023 12:00 PM

→ Steady state cont. extraction in well-stirred vessel

Assumptions:
- Negligible coalescence
- Dispersion phase is small : "lean"
- Feed drops of identical size ; drop breakage is binary & equal
- Conc. changes in drops by diffusion
- Cont. phase (lean no solute) has near zero conc. & changes negligibly due to MT

Solute diffuses out of the continuous phase + → need surface flux (rate @ which solid diffuses out)

Assume you know the age of the drop, & the average conc. inside the droplet

size of the drop

→ Smaller droplets take longer time

→ Infinite breakage rate model → breakup occurs immediately

→ Steady state drop-size distribution { Generations of droplets { 0th → $q_0$ enters (w/o breakup just transferring solute) → if these escaped quickly then there is not much conc. difference/loss in these drops
1st → $q_0$ breaks in $q_1$
2nd → $q_1$ breaks into further

MC → generate random no. & let it play w/ time (Markov chain

Aggregation Model (Coalescence) → blending interior of particles

↳ just grouping together

Aggregation frequency: $a(x,y)$ → fraction of particle pairs $(x,y)$ that aggregate per unit time

$$a(x,y) = a(y,x)$$

Only aggregation (No breakup) : $n(v,t)$ ; $v$ → particle volume

+ We care about all particles that can come together to form agglomerate

↳ Population balance = Mass balance

Total w.r.t dN

Laplace transform → for when you want to deconvolute two integral expression

## Dispersed phase Mixing

Coalescence & breakup occurs

## Cell Mass Model

$$\frac{\partial n(m,t)}{\partial t} + \frac{\partial}{\partial m}[\dot{M} n] = \underbrace{-I(m,s)\, n(m,t)}_{\text{Change in no density (due to division)}} + 2\int_m^\infty I(m',s)\, p(m|m',s)\, n(m',t) \quad \because \int_0^{m'} p(m|m',s)\, dm = 1$$

These two events are the same $\rightarrow p(m|m',s) = p(m'-m|m',s)$

$$\int_0^{m'} m\, p(m|m',s)\, dm = \frac{1}{2} m'$$

Cell division does not change overall mass

Total no. of cells $\rightarrow N(t) = \int_0^\infty n(m,t)\, dm$

$$\Rightarrow \frac{dN}{dt} + \left[\dot{M}(m,s)\, n(m,t)\, dm\right]_0^\infty = -\int_0^\infty I(m,s)\, n(m,t)\, dm + 2\int_0^\infty dm \int_m^\infty dm'\, I(m',s)\, p(m|m',s)\, n(m',t)\, dm'$$

$$\frac{dN}{dt} = -\int_0^\infty p(m,s)\, n(m,t)\, dm + 2\int_0^\infty dm' \int_0^{m'} dm\, I(m',s)\, n(m',t)\, p(m|m',s) = \int_0^\infty p(m,s)\, n(m,t)\, dm$$

$$\frac{\partial n(x,t)}{\partial t} = \frac{1}{2}\int_0^x a(x-y,y)\, n(x-y,t)\, n(y,t)\, dy - \int_0^\infty a(x,y)\, n(x,t)\, n(y,t)\, dy$$

$$\frac{dN}{dt} = \frac{1}{2}\int_0^\infty dx \int a(x-y,y)\, n(x-y,t)\, n(y,t)\, dy - \int_0^\infty dx \int_0^\infty dy\, a(x,y)\, n(x,t)\, n(y,t)$$

$$= \frac{1}{2}\int_0^\infty dy \int a(x-y,y)\, n(x-y,t)\, n(y,t)\, dx \qquad x-y = u,\ dx = du$$

$$= \frac{1}{2}\int_0^\infty dy \int_0^\infty dx\, a(x,y)\, n(y,t)\, n(x,t)\, dy$$

$n(\underset{\sim}{x}, \underset{\sim}{r}, t) \Rightarrow \frac{\partial n}{\partial t} + \underset{\sim}{\nabla}\cdot\dot{\underset{\sim}{x}}(\underset{\sim}{x},\underset{\sim}{r}) + \underset{\sim}{\nabla}_{\underset{\sim}{r}}\cdot\dot{\underset{\sim}{r}}\, n$

Monte Carlo simulations: if you know the distribution → You can generate random variables to model that distribution

**Precipitation Reaction:** occurs because system wants to reach equilibrium

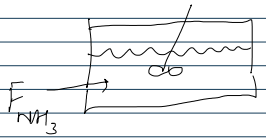
→ Solubility products is exceeded (leads to supersaturation)

→ Mixing of two liquids giving rise to chemical reactions b/w ionic species

} You have to play w/

1D model, $L, t$ ← How long is the crystal grown

↑ characteristic length




$$NH_3^{(aq)} \overset{K_b}{\rightleftharpoons} NH_4^+ + OH^-$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3^{aq}]}$$

$$M^{a+} + \alpha OH^- \rightarrow \underbrace{M(OH)_\alpha}_{\text{precipitation reaction}}$$

$$K_s = [M^{a+*}][OH^{-*}]^\alpha \qquad * \text{ denotes equilibrium}$$

$K_s = [L^{n-}]^* [M^{m+}]^*$ * denotes equilibrium

↳ solubility product

supersaturation

↳ $\Lambda = \frac{[M]}{C^2}$

$K_w = 10^{-14} = [H^+][OH^-] = [H^+]C = K_w$

↑
ionic product of water

Principle of electroneutrality: Anion $A^-$ remains at its original level of $[M]_0$ (since it does not participate in the reaction)

↑ precipitate particles are consuming $OH^-$

↑ Have to donate $OH^-$ to maintain $K_w$

$L$ → rate of growth of particle

$M$ → precipitate

Lecture MT2 review

Tuesday, 18 April 2023 12:09 PM

You only always know the local reaction rate (rate assuming you know the local concentration)

} Idea of effectiveness factor ↓ to get actual reaction rate

($\eta$ corrects/accounts for transport effects)

Coarse grain model (Zurkely?)

Zukaly

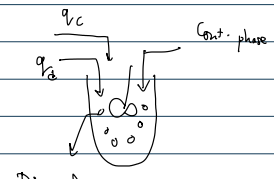
$\eta$ → changes along the axial reactor

$$D\frac{d^2c}{dx^2} - V\frac{dc}{dx} - k'C = 0$$

# Optimization of Chemical Reactors

Optimality Principle

Fixing output conc. & minimizing the residence time

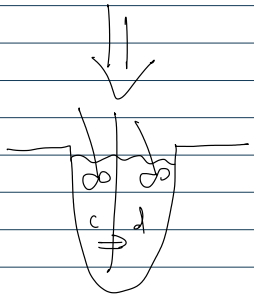



Reaction occuring in cont. phase (includes MT & reaction)

$\frac{d}{dt}[V_c c] = q_c c_f - q_c c - r_{rxn\ term}$ } for cont. phase

$\frac{d}{dt}[V_d c'] = q_d c'_f - q_d c' - MT$ to cont. phase } for dispersed phase

This sort of phase analysis is good only if you have very intense / fast mixing




Population balances are supposed to be a better approximate of this phase mixing.

Pop. balance:

$\int_0^{c_0} n\, dc_1$

$\bar{h} = \omega N_n$

$h^f = \omega\, dc_1 \iint_{c_1+c_2=2c} n(c_1,t)\, n(c_2,t)\, dc_2$

↳ Integrating over an area.
(But only integrating once (Not over a surface)

# / time — Number density

$N|(c)$ → reciprocal of all internal variable

density in conc-space

Closed book:

Reaction invariance → where is it coming from (from Stoich)

Fresh alternative: $A_{\sim} x_{\sim} = b_{\sim}$

$A_{\sim} x_{\sim} = 0$ → If this has non-zero solution

$$\underset{\sim}{A}^T \underset{\sim}{Q} = 0$$